

HOSPIHEALTH



DETAILED PROJECT REPORT FOR CANCER HOSPITAL, KOTA-RAJASTHAN A PROJECT OF INDORAMA INSTITUTE OF MEDICAL SCIENCES PRIVATE LIMITED

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November 2025

To,
Mr. Arjun Agarwal
Indorama Institute of Medical Sciences
Kota, Rajasthan, India

Sir,

We are pleased to submit our Technical & Financial Feasibility Report for Proposed Kota Cancer Centre. The report has been prepared after site visits to Cancer Hospitals in Jaipur and Kota and all other districts Hadoti Region that is Baran District and Bundi District along with meetings doctors and other stake holders. The information thus collected has been analysed for conceptualization of the project. The same data from market research is used for computation of the financial feasibility.

The report is confidential and prepared solely for the purpose of the said assignment, it cannot be circulated, quoted, disclosed, or distributed to any other party without our prior written consent for any other purpose.

Regards,

Dr. Vinay Kothari
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I. INTRODUCTION

➤ PROJECT BACKGROUND

- Kota with Population of over 1.5 million is third largest city of Rajasthan after Jaipur and Jodhpur. It is part of the Hadoti region that includes Baran District, Bundi District and Jahalwar District apart from Kota District.
- Known for its Kota stone industry with multiple quarries and related industries.
- Known across India for its Coaching Industry for various competitive entrance exams like JEE, NEET, IAS, etc.
- It's a major commodity hub for cotton, millets, oil seeds, wheat, etc.
- In addition, Kota and its neighbouring regions has thermal plants, power plants, nuclear plant, fertiliser manufacturing plants, textile weaving industry, etc.
- A Corporate business house having its ancestral roots from Kota plans to commemorate the promoter's Grand Mother through development of a World class Indorama Institute of Medical Sciences. It will be a non-profit organization which will provide quality cancer-care services to the population of entire Hadoti region and entire community who seek quality cancer care at an affordable cost.

- **HOSPITAL MANAGEMENT CONSULTANTS**

- The Promoters sought to understand the Technical & Financial Feasibility of the proposed project through a scientifically performed market study with development of detailed project report (DPR)

23 YEARS

250
PROJECTS

8 MEDICAL
COLLEGES

12.0
MILLION SQ
FT DESIGN



INITIATIVES
IIHH
VIVANS
HOSPITALS
HEALTHCARE
KNOWLEDGE
FEST

PPP &
CSR
STRATEGY

- The task of Conceptualization of the project with its Technical Feasibility Study has been assigned to HospiHealth Consultants India Private Limited, Mumbai, India's leading Hospital and Healthcare Planning Consultants.
- HospiHealth is a knowledge consortium of professionals qualified in hospital and healthcare facility planning, architecture, project & construction management, biomedical engineering, hospital operations management, hospital finance analysts, etc.
- The team has immense knowledge and experience of completing more than 250 hospital & healthcare projects in India and abroad.
- Some of the esteemed clients are as follows:

✓ Hospital Planner & Medico-Technical Consultants, MOHFW Govt. of MP	✓ Breach Candy Hospital, Mumbai
✓ Hospital Planning Consultants, National Health Mission, Maharashtra	✓ Saifee Hospital, Mumbai
✓ Health Authority of Abu-Dhabi	✓ Wokhardt Hospitals (Navi-Mumbai / Rajkot / Bhopal / Nasik)
✓ Al-Rajhi International Clinic, Alkhobar, KSA	✓ Unity Hospital, Mangalore
✓ Chirayu Medical College and Allied Institutes, Bhopal.	✓ Indiana Hospital, Mangalore
✓ D.Y. Patil Medical College & Hospital, Pune	✓ Columbia Asia
	✓ Apollo Hospitals
	✓ Woodland Hospital, Kolkata

➤ **PROJECT CONCEPT**

Location: Swami Vivekanand Nagar, Kota

- ✓ Strategic location along the river.
- ✓ Fast developing with institutional and residential complexes coming up in the area
- ✓ Patients from both old city and new developing area can easily travel to site
- ✓ Accessible to patients from entire Hadoti region
- ✓ **Plot Size:** 15000 Sq. Meters, good for creating 100 bed hospital with scope for expansion and creating amenities with sufficient parking

Based on the market research Hospihealth Consultants will conceptualize the proposed project on a “Social Business Model” which should take care of “Mind, Body & Soul” using Modern Medicine with Humane Touch in highly conducive aesthetic environment. This will be done through development of a **Cancer Hospital** to provide world class healthcare services from preventive to curative care i.e., “**Total Care**” to the community at an affordable cost.

Indorama Institute will comprise of all type of Cancer & Cancer Superspeciality services.

The Hospital will be developed in line with corporate hospitals so as to provide Cancer Services professionally and ethically to develop justifiable productivity for sustainable financial viability of the project.

➤ TECHNICAL FEASIBILITY STUDY

GOAL

- Develop a Cancer Institute “Par Excellence”
- Provide all services under one roof “Total Care”

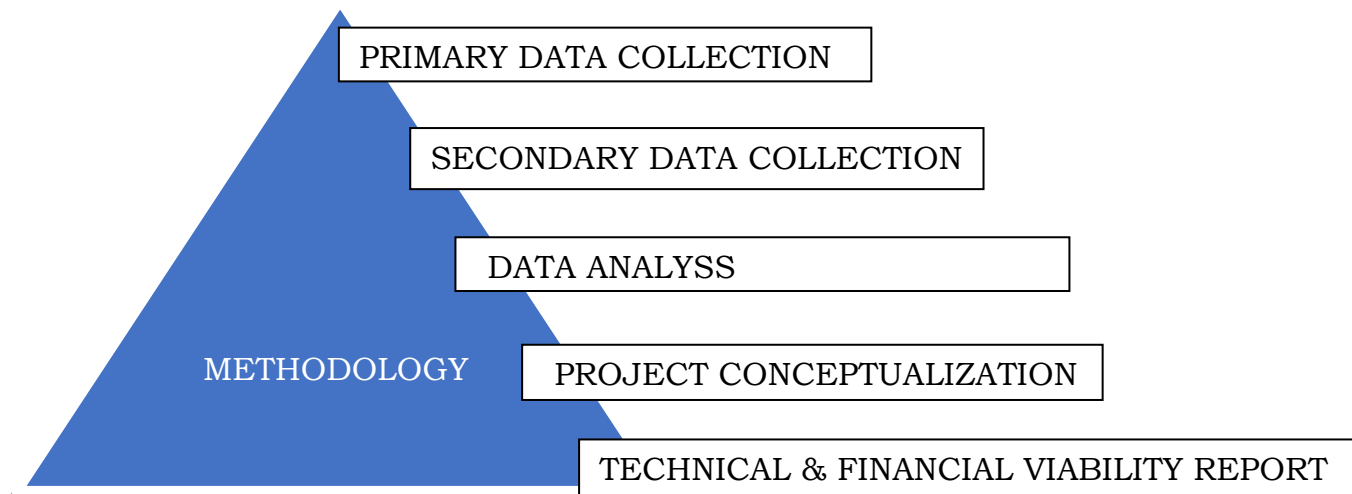
ASSESSMENT OBJECTIVE FOR THE STUDY – MARKET RESEARCH

- Definition of service areas for the proposed hospital
- Evaluating present level of healthcare services
- Exploring facility and service mix for the hospital to have a justifiable productivity
- Provide information like salary details, tariff details, productivity details, etc. for computation of the financial feasibility

PROJECT OBJECTIVES

- Bridge the deficiency of Quality Cancer-care Services in the region
- Provide cost effective “Patient Centric” quality healthcare
- Deploy Latest Medical Technology
- Conducive working environment
- Employ competent staff and offering scope for development
- To explore opportunities to promote Research
- To be Financially Self Sustainable
- Professional and Ethical Management

➤ **METHODOLOGY:**



➤ **PRIMARY DATA COLLECTION**

- The primary data has been collected through personal visits to healthcare facilities in and around service area
- Survey of **medical consultants and other stakeholders** practicing in the area through informal interviews to understand the local healthcare market, drainage area, deficiency, payment mechanism, etc
- Survey of **healthcare facilities**, in the region to get an indication of the level of healthcare provided and the infrastructure available. Productivity indicators, tariff structure etc.

Doctors

- Visit for interview
- Data collection through informal

Hospitals

Site visit to hospitals.
Data collection - productivity,

Primary Service Area Visits:

- Kota
- Bundi
- Baran

Secondary Data Collection - sources:

- Directorate of Census
- National Cancer Registry
- Health Department Reports

- Rajasthan Government Health Policy
- National Health Policy
- Insurance Companies Websites

- Internet Searches
- Cancer Hospital Bulletins like BMCHRC, Jaipur
- Hospihealth Database

The data thus collected will be is used for:

- Project Conceptualization with development of Facility plan
- Computation of Financial Feasibility.

ENVIRONMENT SCAN

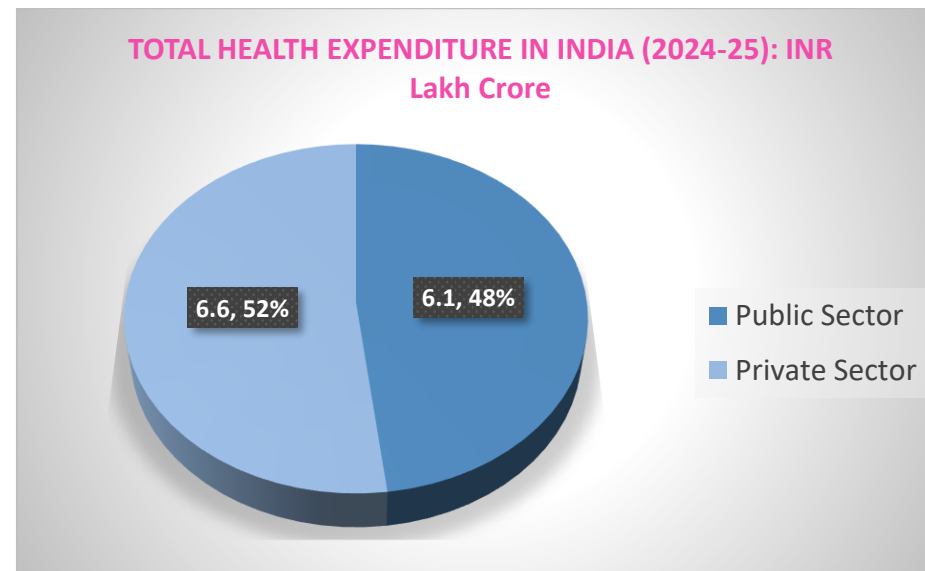
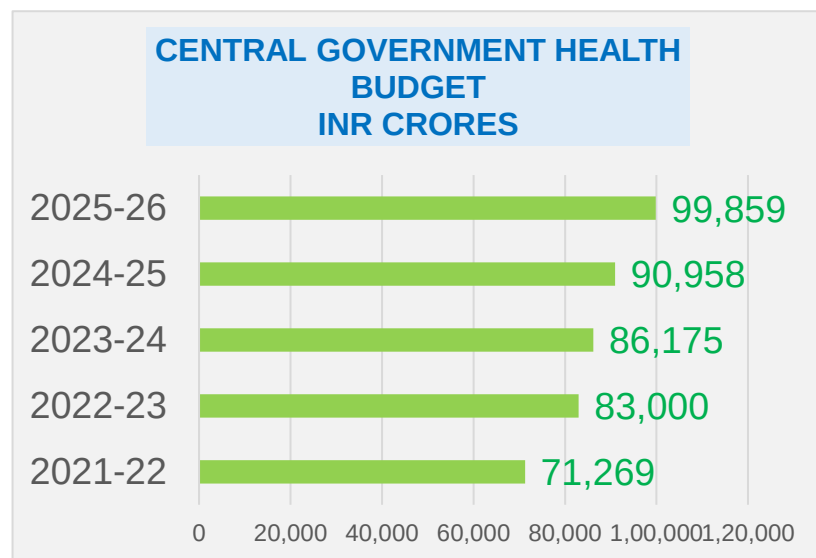
i) INDIAN HEALTHCARE SCENARIO

- Indian healthcare transition from low cost care to high end technology based high cost quality care.
- Government of India launched The National Rural Health Mission (NRHM) in April 2005, to provide accessible, affordable and quality health care to the rural population and adding hospital bed capacity. Goal is to improve all the healthcare indices.
- National Urban Health Mission (NUHM) was launched in May 2013 as a sub-mission to NRHM. The focus of NUHM is to improve healthcare accessibility to urban poor and to reduce their out of pocket expenses by strengthening the existing health care service delivery system
- Health is a fundamental right of the citizens hence it is the duty of the government to provide it. It is being achieved through introduction of **PMJAY Scheme (Pradhan Mantri Jan Arogya Yojana) in Sept 2018** wherein all the **BPL sections** of the society are insured for a **health-cover of Rs. 500,000** to get treatment in public and empanelled private sector hospital.

Major Policy Changes in last 5 years:

- Up-gradation of district hospitals into medical colleges, the current number of total medical college is 785
- Intake capacity at MBBS level has been increased to 1,18,620
- Under Pradhan Mantri Swasthya Suraksha Yojana (PMSSY) there is augmentation of tertiary healthcare capacity, medical education and research in underserved areas of the country.
- **On Oct 2024, all senior citizens aged 70 and above, regardless of their socioeconomic status** are covered under PMJAY. A family floater health cover of up to ₹5 lakh per year for secondary and tertiary care is provided. For BPL population of senior citizens, this is in addition to 5.0 lakh cover already provided.
- Under PMSSY AIIMS like institutions are being built and Government Medical Colleges are upgraded by setting-up Super Speciality Blocks.
- Recent initiative of allowing 200 bed plus hospitals to conduct post graduate courses
- Free dialysis scheme, benefited roughly 2.5 million people.

➤ **HEALTH EXPENDITURE IN INDIA:**

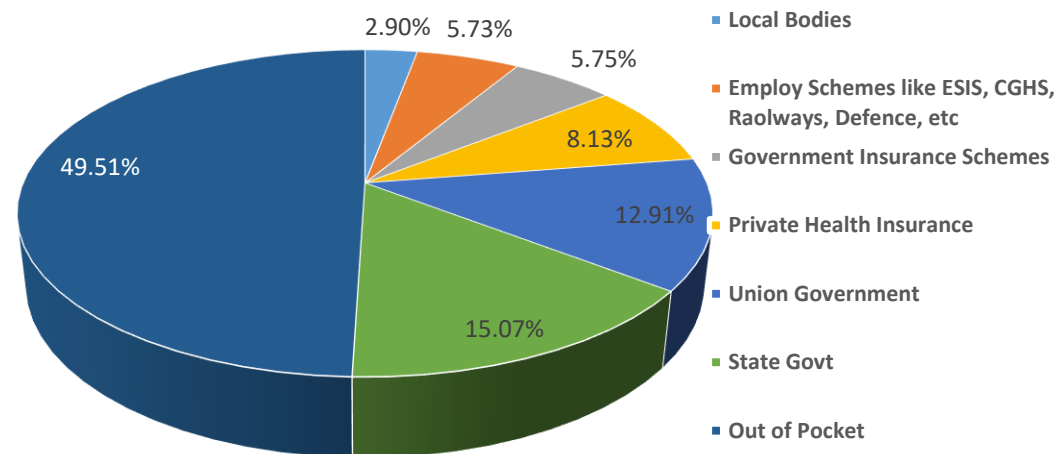


Source: National Economic Survey estimates for 2024-25 presented in parliament

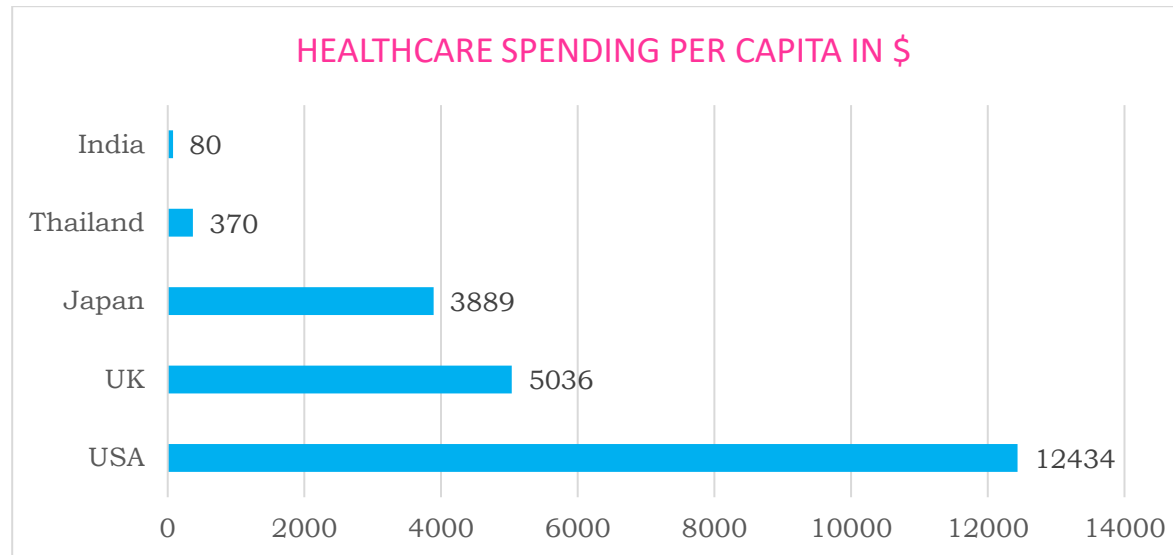
- The **Ministry of Health & Family Welfare** has been allocated **₹99,859 crore** for **FY 2025-26**, up from ₹89,974 crore in FY 2024-25 (RE), reflecting an approximately **11% year-on-year increase**.
- Total spending inclusive of public & private sector is about 3.8% of GDP
- **India's hospital industry** is estimated at around **USD 136.6 billion in 2024**, with projections to reach **USD 264.8 billion by 2033**

- **Public sector healthcare spending**—It was **6.1 lakh Crore in 2024-25** up from 3.2 lakh crore in 2020-21, a CAGR of 18%. Public sector spending comprise of budget of Government of India, all state governments, all major public sector ministries & corporation like Defence, Railways, ESIS, etc
- **Private Sector** (Out of pocket expenditure) spending is around 52% that is approximately **6.6 lakh Crore**. Thus, total expenditure is more than Rs.12.7 lakh crore

COMPOSITION OF INDIAN HEALTHCARE EXPENDITURE

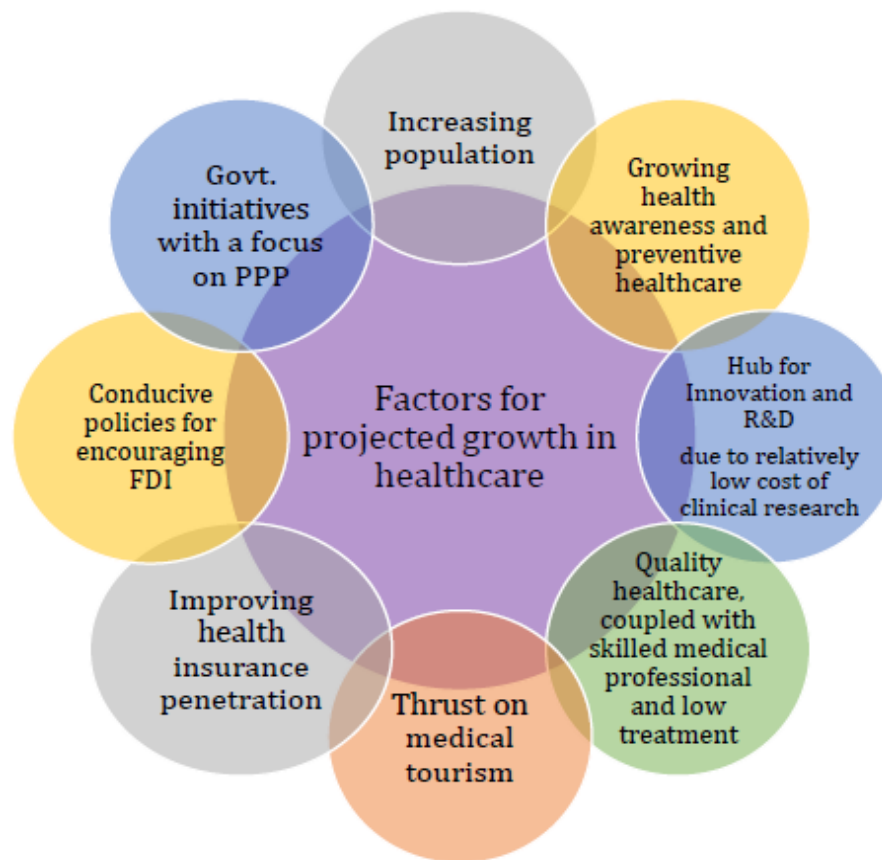


- **Per Capital Healthcare Expenditure** of India is one of the lowest in the world, even less than countries like Thailand.



India is a fastest developing global economy. The impact is on healthcare too with huge healthcare potential for development i.e. serving 1.46 billion population. *Due to its cost benefit advantage and capability of providing World class healthcare services, India is gradually becoming the most attractive International Healthcare tourism destination.*

Patients from across the globe have already started coming for Joint Replacement, Cancer Treatment, Cosmetic Procedures, Dental & Ophthalmic procedures, Treatment of chronic diseases through alternative therapies, etc. Soon there will be a rush of international patients from across Europe and may be even USA for specialities or procedures having high waiting time.



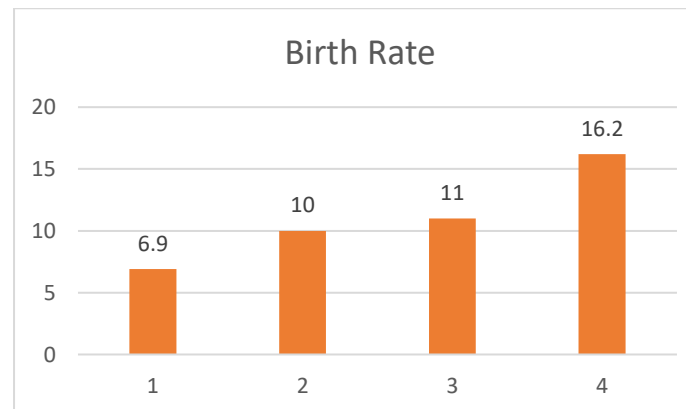
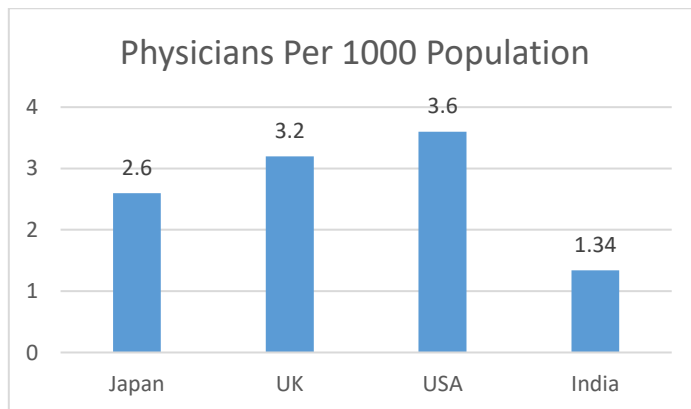
However, we are **yet to catch up with other developing nations on key indicators** such as life expectancy, IMR, MMR, Birth rate etc... **To improve its performance** and meet India's future healthcare **needs**, the government and industry need to take actions that can increase **investment and employment in healthcare**.

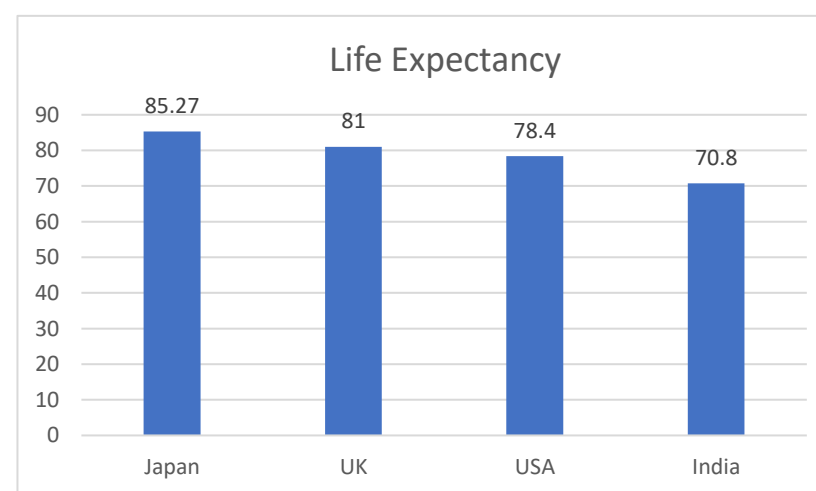
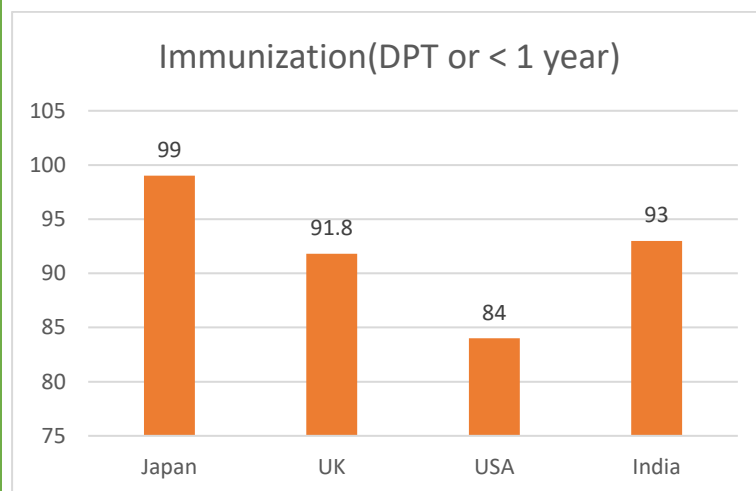
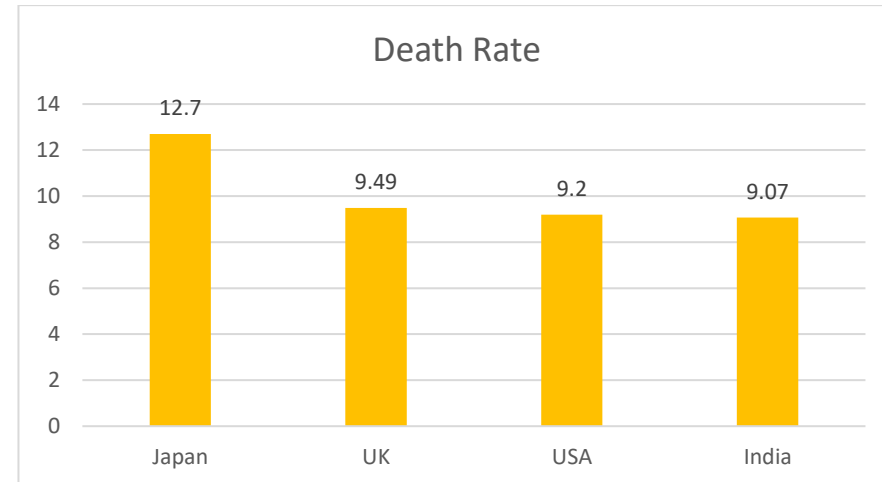
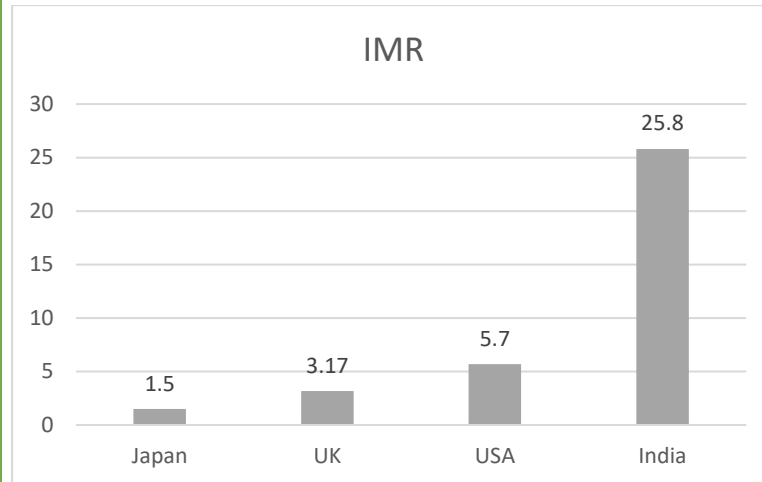
Healthcare infrastructure in India is improving fast and now coming close to other middle-income countries (like China, Brazil, Thailand, South Africa and Korea)

✓ **Healthcare Indicators**

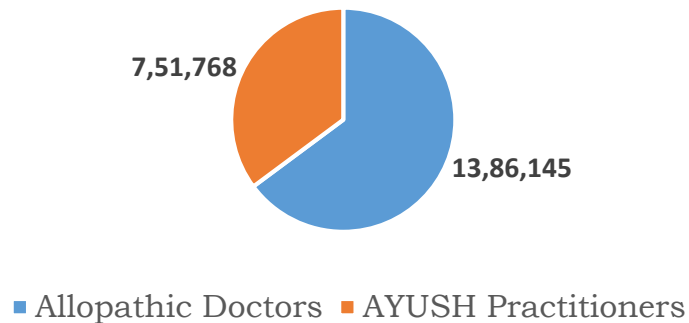
Country	Japan	UK	USA	India
Physicians /1000	2.60	3.20	3.60	1.34
Birth Rate	6.9	10	11	16.2
IMR	1.50	3.17	5.70	25.80
Death Rate	12.7	9.49	9.20	9.07
Life Expectancy	85.27	81.00	78.40	70.8
Immunization(DPT or < 1 year)	99.00	91.80	84.00	93.00

Source: IMR: macrorends.net 2019-20 / www.vox.com, who.int / statista.com/oecd /





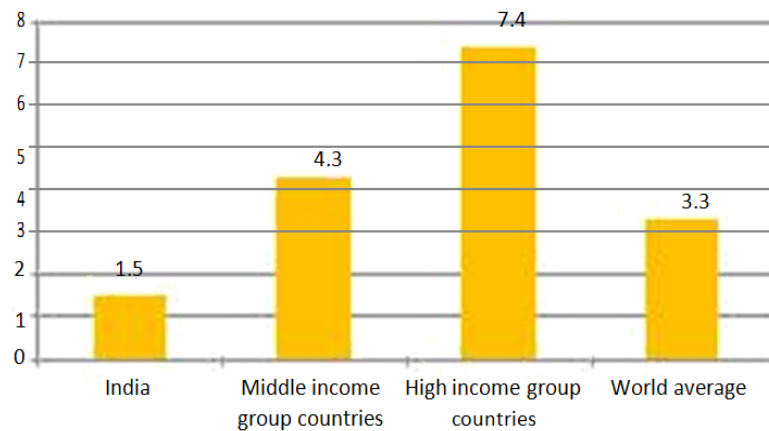
Ratio of Allopathic Doctors to AYUSH Doctors



Doctor to Population Status in India

- ✓ Total number of registered allopathic doctors in India: 13,86,150 India doctor population ratio: 0.95:1000 (Registered)
- ✓ India doctor population ratio: 1.5 inclusive of ISM
- ✓ WHO Recommendation: 1 per 1000
- ✓ Number of MBBS seats in India is approximately 118,620. In 2018, it was about 60000.
- ✓ Mal-distribution of doctors: more in urban area as compared to rural

Bed per 1000 population



Hospital Bed Status in India

- **Total number of hospital beds in India:** 2.1 million (Public Hospitals / Private Hospitals / Medical Colleges)
- ✓ **WHO Norm:** 3 beds / 1000 population
- ✓ **World Average:** 3.3 beds / 1000 population
- ✓ **Indian Average:** 1.5 beds/ population
- ✓ **Mal-distribution** of beds between urban and rural areas
- ✓ **Deficiency** of **tertiary care** like Cardiology, Neurology, **Cancer Care**, Joint Replacements. Trauma Care, etc

HEALTHCARE SCENARIO OF THE COUNTRY CAN BE SUMMED UP AS FOLLOWS:

a) Grossly inadequate healthcare infrastructure

- low bed: population ratio - there are about 21 lakh hospital beds in India, or about 1.5 beds per 1000 people; to reach a ratio of 2 beds per 1000 as per new health policy, India needs to build another 7.50 lakh hospital beds;
- Excessive overlap across primary, secondary and tertiary care
- Lack of standardization/ effective regulation leading to wide disparities in quality, availability and cost of services; this disparity is again supported by varied costs of manpower and consumer's purchasing power
- Skewed towards urban populace

b) Increased demand due to:

- Socio-economic development, increased literacy rate with increase healthcare awareness
- Increased life expectancy leading to increase in geriatric population
- Life style diseases of elite i.e NCD's (Non-communicable diseases)
- Increase penetration of health insurance

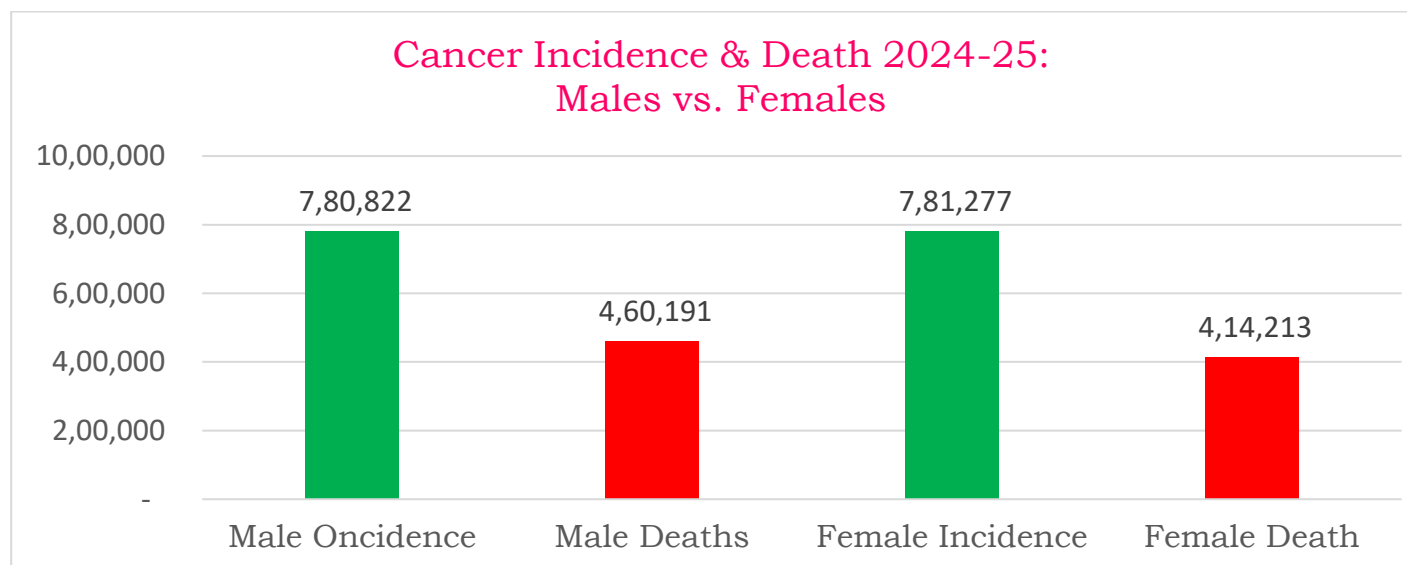
c) Negligible understanding of healthcare economics

- Lack of adequate corporatization
- Obsession with beds and hi-tech equipment
- Confusion about the meaning of "not-for-profit"
- Focus on occupancy rather than throughput

CANCER SCENARIO: INDIA VS RAJASTHAN

Cancer Incidence Rate (2024):

- Cancer is the **fourth most leading cause of death in India**
- There is a **Central Cancer Registry in India** which track cancer through 43 cancer registries across India, to track cancer incidence with major sites of cancer and age specific incidences
- Rajasthan has 1 such registry in Bikaner, however Bhagwan Mahavir Cancer Hospital & Research Centre tracks cancer based on its high volumes of patients.



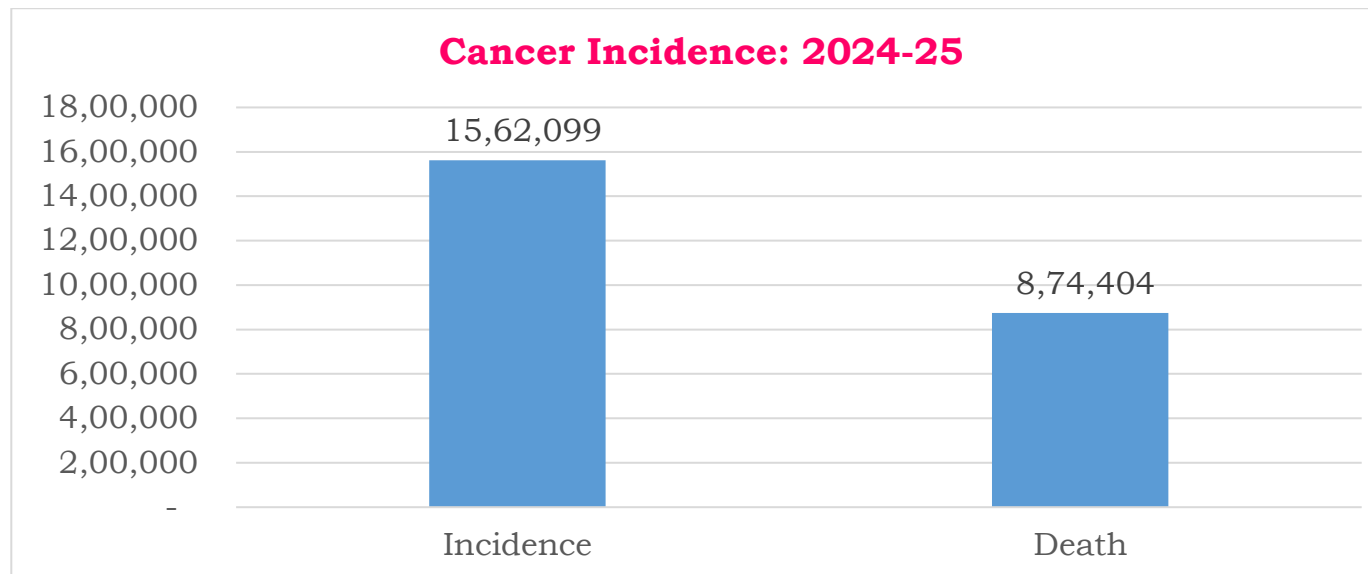
- According to a comprehensive analysis of data from 43 population-based cancer registries in India had about **1,562,099 new cancer cases estimated for 2024-25** as against 1.39 million in 2020 (GLOBOCON).
- The incidence of cancer cases is estimated to increase by 12.8 per cent in 2025 as compared to 2020
- The **lifetime risk** of developing cancer in India is about **11%**.
- **Male Cancer Incidence (2024-25):** 780,822
- **Female Cancer Incidence (2024-25) :** 781,277
- **Female Cancer Incidence is higher than males**
- Most **common male cancers:** Oral cavity with pharynx, lung and prostate cancers
- Most **common female cancers:** Breast Cancer, Cervical cancer and ovarian cancers
- **0-14 age group Leukaemia** accounted for highest cancer incidence

Cancer Prevalence Rate:

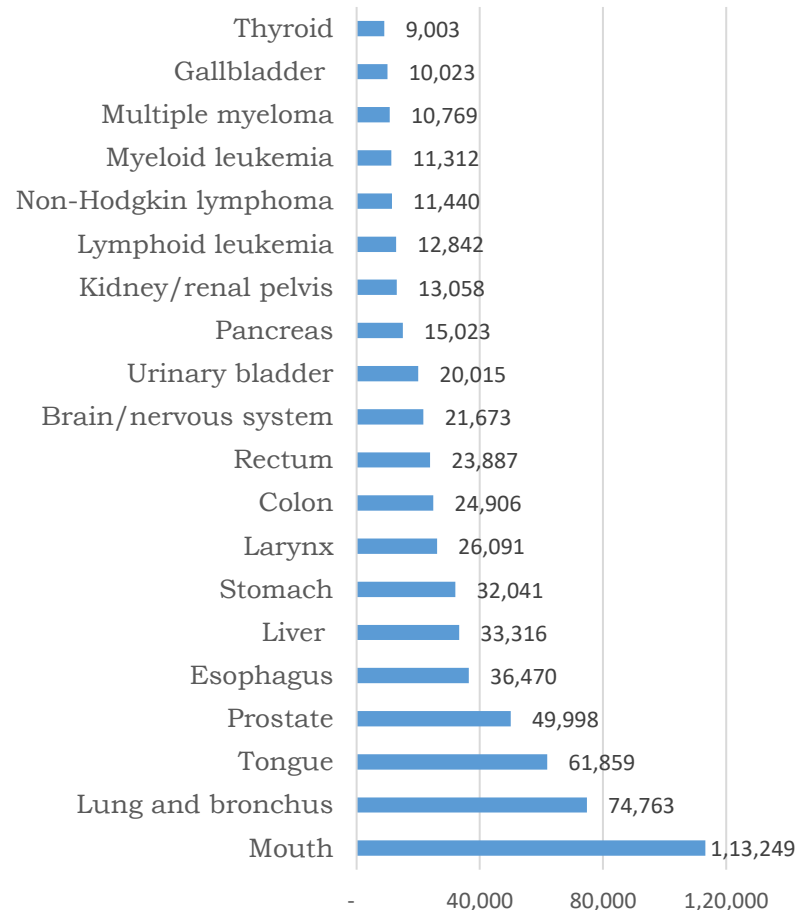
As per a report from ICMR cancer prevalence in India is **about 2.5 million**. It includes pre-existing cancer cases and newly reported 1,59 million cases

Cancer Death Rate (2024-25): India

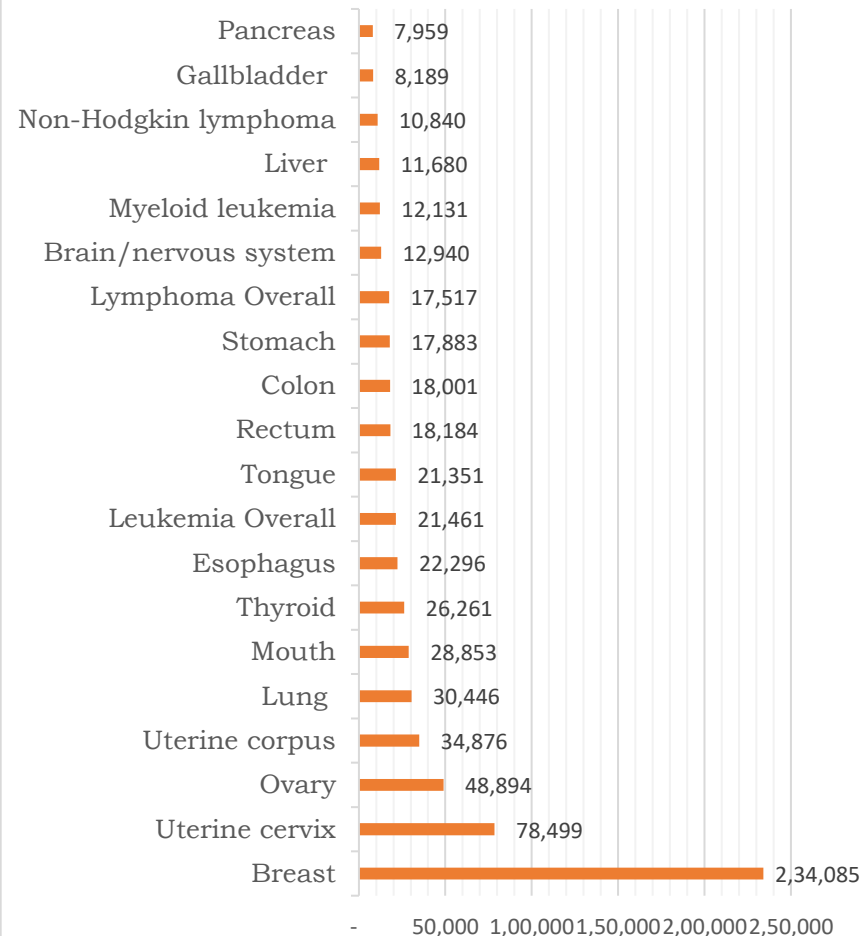
- Approximately **8,74,404 death** were reported due to cancer **in 2024 that is 56% of the total 15,62,099 new cases**
- **Males** accounted for about **4,60,191 death** as compared to females accounting for about **4,14,213 death**
- **Males have higher risk of cancer death** in spite of relatively lower incidence.
- **However females have lower risk of death** in spite of higher incidence



MALE CANCER INCIDENCE - TOP 20 LOCATIONS



FEMALE CANCER INCIDENCE - TOP 20 LOCATIONS

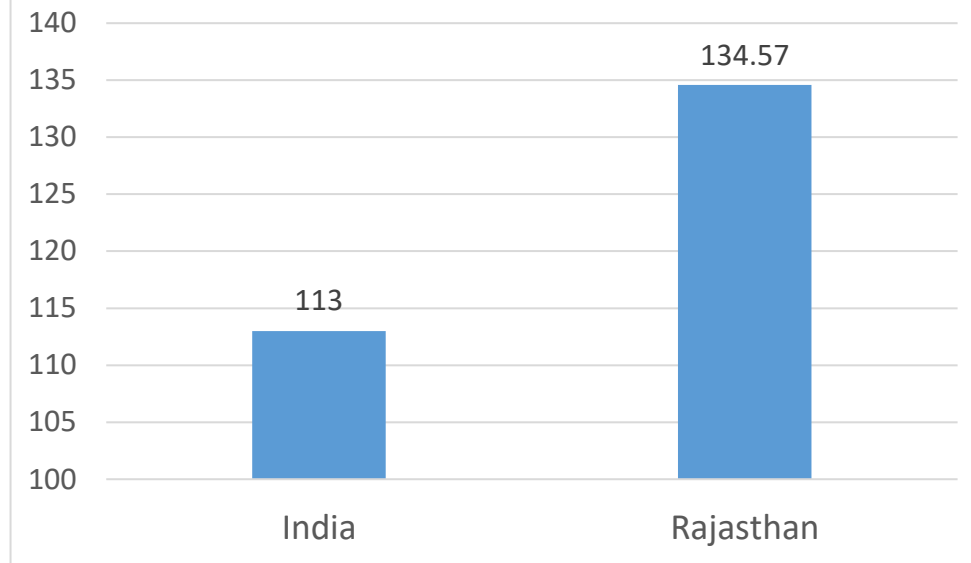


Source: JAMA (National Cancer Registry Programme Investigator Group)

Cancer Incidence Rate (2024-25): Rajasthan

- A report from Bhagwan Mahavir Cancer Hospital & Research Centre, Jaipur, reported to Indian Council of Medical Research Cancer registry projects the incidence rate at 134.57 per lakh population compared to the national average of 113 per 100,000 population
- Rajasthan Population is 85.4 million thus based on the incidence of **134.57 per lakh** population total number of cancer patients in **Rajasthan will be about 1,14,922.8**

Incidence per Lakh Population



- There are major geographic variations in cancer incidence in Rajasthan with Jaipur, Ajmer Bikaner and Dausa region accounting for maximum incidence
- Jaipur has the highest cancer incidence rate, with 332 cases per 100,000 population

- Ajmer : 256 per 100,000 Population
- Jhunjhunu : 235 per 100,000 Population
- Dausa : 225 per 100,000 Population
- Most prevalent Cancers: cancers of the lip, oral cavity, and pharynx 26-32% approximately, digestive organs 15% approximately, breast 11-12% approximately, respiratory and intrathoracic organs 10-11% approximately and female genital organs 8-9% approximately.
- High incidence is attributed to Tobacco, Alcohol, Pollution & Pesticides

➤ **Strengthening of the health care systems:**

Rajasthan government has embarked on a mission to revolutionize healthcare delivery across the state. Recognizing the importance of accessible and affordable healthcare for all its citizens, the government has implemented a comprehensive health scheme aimed at providing quality medical services, improving infrastructure, and ensuring financial protection. Some of the health schemes followed by the government of Rajasthan are:

KOTA DISTRICT : DEMOGRAPHIC PROFILE

Population 19.51
Lakhs
2025 - 24.90 Lakhs

Literacy Rate
82.80%

PROPOSED
INDORAMA
INSTITUTE of
MEDICAL
SCIENCES

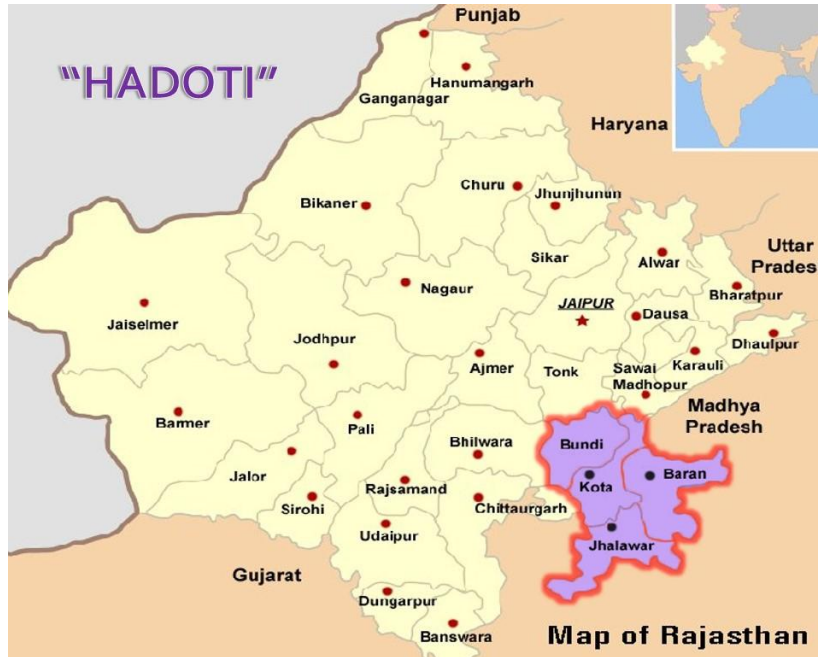


Area
5217 sq. km.

Sex Ratio
M:F
1000 : 895

2020-21
Per capita income
Rs. 1.25 Lakhs
2025
Ra. 1.85 Lakhs

HADOTI REGION:



Hadoti Region comprise of 4 districts with approximate current population of about **7 million**.

1. Kota District : Population 2.49 Million
2. Baran District: Population 1.42 Million
3. Bundi District: Population 1.3 Million
4. Jhalawar District: Population 1.72 Million

(All population figures are estimates based on 2011 census)

Kota is a fast-growing city in all the aspects like Residential Complexes, Shopping Malls, Educational Institutes, Recreational activities, etc., however its healthcare infrastructure to provide high quality healthcare is highly deficient. This augurs well for the promoters to develop a top line corporate hospital providing international level quality healthcare.

Economic Status:

- Kota is one of the industrial hubs in northern India, with chemical, cement, engineering, and power plants. As per the DIC records there are more than 12000 units in Kota, 6000 in Bundi, 1000 in Barah and 11000 in Jhalawar.
- **Rajasthan's** 2025 estimated per capita income is about **INR 1.85 lakhs**

➤ **COMMERCIAL LANDSCAPE**

➤ **Coaching Industry:**

- Kota has more than 48 coaching classes with nationally recognised names like Bansal Coaching Pvt. Ltd, Allen, Carrier Point, etc.
- Current industry size is between 3500-4000 crores with students between 1.0-1.25 lakhs
- Few years back it used to be between 8000-10000 crores with more than 2.5 lakh students
- Coaching classes drives Hostel business, restaurant business, transportations business and such other activities, now with the slump most of the hostels have less than 50% occupancy

➤ **Power Plants:** Kota Region has four power plants within 50 km radius

- Rajasthan atomic power plant at Rawatbhata



- Coal based Kota Thermal Power plant
- Anta Gas Power plant in Baran District
- Jawahar Sagar Hydraulic Power plant Jhalawar
- Surya Chambal Power Plant in Rangpur (biomass)

Power plants show annual growth of 15–20% due to their strategic locations.

➤ **Kota Stones:**

Hadoti region is known across India fine-grained variety of limestone, Kota Stone which comes with rich greenish-blue and brown colours. It is high quality stone which is tough, non-water-absorbent, non-slip, and non-porous. The region has high number of quarries with many industrial units for polishing Kota stone, making tiles, etc.



Commodity Market: Kota is famous for cotton, oil seeds, wheat. etc. It is one of the largest oil seed market of the country

➤ **Chambal Fertilizers:** Asia's biggest private sector fertilizer manufacturer in Kota

➤ **Kota Doriya:**

Kota is known for its famous Kota Sarees having many weaving units

➤ **Kachori:**

An Indian snack is famous from Kota having hundreds of small units. It is even exported in large number.

➤ **City Landscape:**

Kota has large number of commercial establishments like Hotels, Malls, Office complexes and housing industry. Real state sector is booming.

➤ **Accessibility:**

Road: It is well connected to Jaipur, Delhi, MP, Haryana, etc. by road

Rail: well-connected by rail from Rajasthan, Gujarat, Maharashtra, MP, UP, Haryana and Delhi.

Air: A new International airport is coming up at Kota thus soon it will be well connected by air



➤ **HEALTHCARE STATUS: KOTA**

➤ **PROFILE OF HOSPITALS:**

- **Medical College Hospitals**

- Government Medical College:
- MBS Hospital, attached to Government Medical College
- Sudha Medical College.

- **Major Multispeciality / Superspeciality Hospitals: 100-250 bed**

- Sudha Hospital & Medical Research Centre
- Maitri Hospital
- Kota Heart Institute & General Hospital
- Sriji Multispeciality
- Ethos Hospital
- **Trust Hospital** - Bharat Vikas Parishad Hospital

- **30-100 Bed Private Hospitals**

- Dhakar Hospital
- Ramkrishna Hospital

➤ Radhakrishna Hospital

➤ Navkar Hospital

- **Small Hospitals & Nursing Homes –**

Approximately 50 such hospitals are there having 8-30 bed capacity

- Healthcare infrastructure development in Kota could not keep pace with fast urbanization and population growth.
- This has led to major healthcare deficiency in the area especially for high end diagnostic services & super speciality services
- Entire belt is served by mainly few mediocre hospitals that is 3 Multispeciality Hospitals, 1 trust hospital and 3 Teaching hospitals, 2 corporate like hospitals that is Sriji Multispeciality Hospital and Ethos Hospital, rest are either mediocre 50-100 bed hospitals or highly primitive small hospitals and nursing homes
- Quality of services are poor except in Ethos Hospital.
- The area has witnessed high population growth and is further growing south toward Jhalawar and east towards Baran.
- This will require development of high-quality healthcare infrastructure to serve this population as the current hospital services are much below the desired standards.

Major deficiencies of the area are:

- Large multispeciality hospitals providing all the services under one roof
- High end building infrastructure to give perception of high quality hospital
- High end diagnostic services with “State of Art Technology”
- Low quality of hospital services mainly due to primitive building infrastructure and mediocre level of nursing services
- Not many hospitals with full NABH accreditation
- No Hospital with JCI accreditation
- No major centre for Interventional Radiology
- Lack of Professional & Ethical Services
- **Deficiency of comprehensive Cancer Care Services:**

➤ **DEFINING SERVICE AREAS – DEMAND SUPPLY GAP**

A closer look at 'service area' of the proposed project would enable us to get an understanding of important parameters like potential **population, access to healthcare facilities and need assessment that is deficiency in the current healthcare system and demand supply gap.** Initially, it is necessary to outline the '**primary service area**' of a hospital ie *area from which hospital draws its major share of patients.*

- Hospital service area boundaries may coincide with those of recognized civil divisions but this is not the end.
- Patients avail hospital services irrespective of civil division boundaries.
- **Factors that determine service area are:**
 - ✓ **Type of Disease:** for life threatening diseases like Cancer patients tend to travel long distances to avail healthcare services irrespective of the distance to be travelled
 - ✓ **Branding / Quality:** Reputation of the hospital is important for diseases like cancer.
 - ✓ **Availability of Comprehensive Services along with Doctors:** availability of doctors are important especially for cancer for example cancer patients in India tend to choose **Tata Memorial Cancer Hospital in Mumbai**, being the most reputed cancer hospital in southeast Asia but due to high difficulty in getting appointments, they cannot wait for long

and hence choose alternatives like BMCHRC in Jaipur or Rajiv Gandhi Cancer Hospital in Delhi, etc.

- ✓ **Cost of Treatment:** Due to high cost of cancer treatment patients tend to go to relatively cheaper hospital or the hospitals which are empanelled with government Health Insurance Schemes like CGHS, PMJAY, etc.
- *Cancer treatment is a long process takes weeks to months for the treatment and patient tend to stay in and around facility for long time along with relatives hence **cost of stay with local lodging and boarding** is also one of the defining factor*
- ✓ **Accessibility:** *Highway and railroad patterns determine mode and direction of travel by the patients of an area.*
- ✓ People prefer to go to the nearest hospital offering high quality holistic healthcare at an affordable cost.

➤ **DRAINAGE AREA:**

A study of the drainage area of the existing hospitals, and consultants is the criteria for defining service area.

- Kota district – 60-70%
- Baran, Bundi & Jhalawar: 30-40%

- 60% to 70% patients from Hadoti region go to Jaipur mainly to SMS Hospital and Bhagwan Mahavir Cancer Hospital.
- 5-7% of the high economic strata patients population goes to Delhi or to Mumbai

PRIMARY SERVICE AREA

The area from which the hospital will draw maximum patients.

- **Primary Service area** considered for the project is entire Hadoti Region inclusive of Kota, Bundi, Baran and Jhalawar Districts
- **Population: 6.9 million** approximately
- **Population Growth:** The area is witnessing growth of about **2.4% per year**
- **Economic Profile:**
 - The demographic of the population is changing from small to skyscrapers
 - BPL and Lower Middle class gradually changing to Middle and upper Middle class
 - With the development of high end residential and commercial projects in the area, here is flush of high-end executive and elite class as well
 - The proposal for a new airport has been sanctioned and this will see a major shift in the economy of the city in a positive direction.

We need to understand **availability of cancer hospitals providing all the services under one roof** in and around the region to define primary service area. They are:

- Kota to Indore: 321 Km
- Kota to Jaipur: 252 km
- Kota to Udaipur: 284
- Kota to Delhi: 472 km

All these cities providing cancer care are more than 200 kms away.

Hence if there is a high end Cancer Hospital in Kota providing high quality comprehensive cancer care at affordable cost, the prospects of hospital doing well are very high.

SECONDARY SERVICE AREA:

It is the area from which a proposed facility may draw patients even if they are away from proposed hospital.

Main reason for the patients to come from secondary service area are:

- No Cancer Hospital in the entire region
- Non availability of comprehensive cancer care facility providing high quality cancer care
- Poor Branding
- High cost of treatment

Most of the patients from the entire region are travelling mainly to Jaipur, a small percentage especially from districts of MP and from Chittorgarh to Indore followed by some of upper middle class and elites to Delhi or Mumbai.

Jaipur is having many hospitals providing comprehensive cancer services. The main hospitals that patients prefer:

- **SMS Hospital Jaipur:**

used mainly by BPL and lower middle income group being a totally free government hospital

- **Bhagwan Mahavir Cancer Hospital & Research Centre:**

Most of the affording patients from Hadoti region and its neighbouring district inclusive of neighbouring district of Madhya Pradesh travel to BMCHRC being a trust hospital is relatively cheaper and easily accessible by rail and road.

Districts of Rajasthan

The Six Rajasthan districts **that share border** with the Hadoti regiont having **Population of >10 million** and could be potential target area for proposed hospital at Kota.



- **Bhilwara** borders Bundi and Kota
- **Chittorgarh** borders Kota and Jhalawar districts
- **Pratapgarh** borders Jhalawar district
- **Sawai Madhopur** borders Kota and Bundi districts
- **Tonk** borders Bundi and Kota district
- **Karauli** borders Baran district

Districts of Madhya Pradesh

Six Districts that **share border with Hadoti region** and **2 additional districts that are near Kota** could also be potential targets with a **population of >15 million**.



1. **Mandsaur** borders Jhalawar
2. **Neemuch** borders Jhalawar
3. **Ratlam** borders Jhalawar
4. **AgarMalwa** borders Jhalawar
5. **Rajgarh** borders Jhalawar and Kota
6. **Guna** borders Kota

INSTITUTIONAL SURVEY:

➤ COMPETITIVE LANDSCAPE:

- **5 small cancer care nursing home** basically clinics extended with in-patient services.
 - **3 Onco-surgeons** developed operation theatres to provide oncology services.
 - **2 Oncologists** extended clinic with inpatient services to provide chemotherapy

➤ PUSHPADI CANCER CENTER: Srinathpuram, Kota



- Owner: Dr. Manmohan Agarwal, Onco-surgeon
- Small hospital which is an extended clinic with having 6 beds and operation theatre

➤ **NANDINI HOSPITAL:**

Owner: Dr. Harsh Goyal, Oncologist

Location: Basant Vihar, Kota



- Planned as 30 bed hospital with
- Inpatient Bed - General Ward, twin share room, Single, Room, -
- ICU
- Operation Theatre 2
- X-ray
- Pathology
- Pharmacy
- Physiotherapy
- Only 10 bed are operational

➤ AROGRAM CANCER HOSPITAL: Ganesh Nagar, Kota



Dr. Kaushal Gautam, Ms, M.ch.: Onco-surgeon

- Planned as 20 bed hospital with
- Inpatient Bed - General Ward, twin share room, Single, Room, -
- ICU
- Operation Theatre 2
- X-ray
- Pathology
- Pharmacy
- Only 10 bed are operational

➤ SHREE KALYAN HOSPITAL: R.K. Puram, Kota

Dr. Manish Gautam, Oncologist MD, DNB



SMALL HOSPITALS COMPARATIVE

Name of Hospital	Pushpadi Hospital	Arogyam Hospital	Hadoti Hospital	Srikalyani Hospital	Nandini Hospital
Year of Establishment	All of them developed in last 5 years				
Ownership	All privately owned by doctor				
Total Operational Bed	10	12	10	6	30
ICU	2 Post OP Care	N	N	N	3
Chemotherapy	N	N	N	Y	Y
General Ward	4	6	6	N	10
Sharing Room	4	4	2	N	6
Special Room	2	2	2	NA	4
Operation Theatre	1	1	1	N	2
X-RAY	NA	Y	Y	N	Y
PATH	Only sample collection				

Name of Hospital	Pushpadi Hospital	Arogyam Hospital	Hadoti Hospital	Srikalyani Hospital	Nandini Hospital
Bed Occupancy	30%	50%	50%	NA	NA
	2 Post OP Care	N	N	N	3
General Ward	1200	1500	1200	NA	1000
Sharing Room	2100	2500	1800	NA	NA
Special Room	3000	3000	2400	NA	N
Operation Theatre	1	1	1	N	2
X-RAY	300	300	300	N	Y
PATH	Only sample collection				

	Pushpadi Hospital	Arogyam Hospital	Hadoti Hospital	Srikalyani Hospital	Nandini Hospital
Consulting Doctor	Dr Manmohan Agarwal, Onco-surgeon	Dr. Kaushal Gautam Onco-surgeon	Dr. Akhlak Hussain Onco-surgeon	Dr. Manish Gautam Oncologist	Dr. Harsh Goyal Oncologist
OPD					
Patients per Day	12-15	18-20	8-10	10-12	6-8
Rate	600	600	500	600	600
Chemotherapy per Month	3	5	N	10	8
Average Cost	40,000-70,000				
Surgery / month	Major 5-6 Minor 10	Major 6-8 Minor 18-20	Major 2-3 Minor 12-14	N	N
Rate – Major	30,000	30,000	25,000		
Minor	2000-3000	2000-3000	2000-3000		

- All these doctors are attached to 2-3 hospitals where they do OPD as well as operative or chemotherapy work
- Unique Trend: All the surgeons they tend to do chemotherapy also

- **Government Medical College** Teaching Hospital is having Onco-surgery services having one Onco-surgery Unit – ward. Onco-surgeon uses General Surgery operation theatre

Headed by Dr. Harsh Jangid Onco-surgeon.

OPD: 30 per day. Surgeries – 1 per day

- **MBS Hospital:** Used as an extension of Medical College. It is having a Chemotherapy Unit and a Radiotherapy Unit. Chemotherapy is being given by Radiotherapist.



Radiotherapy Department

- Department headed by Radiotherapist Dr. Kanwar and team comprising of Associate Professor Dr Dadhich and junior doctors (PG students). Dr. Kanwar, is authorized to do chemotherapy as well
- Facility: 30 Bed ward : occupancy is 100%
- **OPD:** 20-25 patients per day
- **Admission:** 7-8 per day
- **Chemotherapy Unit** – 4 bed - 3-4 chemo per day
- Radiotherapy Machine: Cobalt
- Radiotherapy : 5-6 therapies per day

➤ **Sudha Medical College Hospital:**

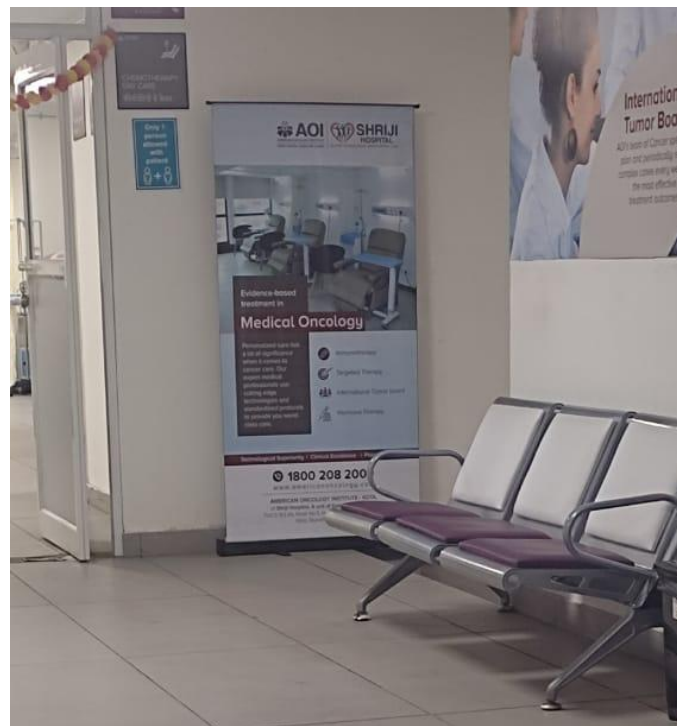
It has an **Oncology ward with full time Oncologist and Radiotherapy Centre** with Linear Accelerator and Brachytherapy. Operation Theatre of Surgical department is being used. Onco-surgeon is Dr. Kaushal Gautam who is also having his own surgical Arogyam Cancer Hospital.

American Oncology Institute (AOI): Sriji Multispeciality Hospital has outsourced their basement area to. They have full time Chemotherapist, Onco-surgeon and Radiotherapist

Facilities:.

Radiotherapy Centre Linear Accelerator & Brachytherapy and PET Scan facility.

AOI uses Medical, Surgical and other diagnostic facilities of Sriji Hospital.



1. Visit to Baran Hospital: Met 4 doctors in Baran



2. Visit to Bundi – Met 4 Doctors



- **BENCHMARKING:**

BHAGWAN MAHAVIR CANCER HOSPITAL & RESEARCH CENTRE (BMCHRC), JAIPUR

There is no top line cancer hospital in Kota or entire Hadoti region. About 70% of the patients from this region goes to Jaipur mainly to Bhagwan Mahavir Cancer Hospital & Research Centre. Hence BMCHRC has been considered for benchmarking as it provides comprehensive care services in all the disciplines and sub disciplines of cancer services.



BMCH CANCER HOSPITAL		
पहला तल	1 level You are here	-1 level वेसमेट तल
OPD-Billing ओपीडी-बिलिंग		G level ग्रे-तल
BPL Counter बीपीएल काउंटर		
Medical OPD मेडिकल ओपीडी		
Radiology Dept. रेडियोलॉजी विभाग		2 level द्वितीय तल
Blood Sample Collection रक्त नमूना संग्रहण		3 level तीसरा तल
OT Complex II ओटी कॉम्प्लेक्स II		4 level चौथा तल
Surgical OPD सर्जिकल ओपीडी		5 level पांचवां तल
Cancer Care Office कैंसर केयर ऑफिस		
Emergency इमरजेंसी		
MICU एमआईसीयू		
Cubical Ward क्यूबिकल वार्ड		6 level छठा तल
Lift/Lobby लिफ्ट/लॉबी		7 level सातवां तल

- **Location:** Jawahar Lal Nehru Marg, Bajaj Nagar Jaipur.
- **Founder:** Mr. Navratan Kothari, KGK Diamonds
- **Bed Size:** 300 bed
- **Year of Establishment:** 1997
- **Ownership:** Trust
- **Plot Area:** 18,537 sq. meters
- **Constructed Area:** 30,000sq meters, 2 new blocks are being built



- It is one of the **only private (trust) cancer hospital in north-western part** of India.
- **Among top five cancer hospitals** in the country.

- Provides high quality comprehensive cancer care services, all under one roof inclusive of Medical Oncology, Surgical oncology and Radiotherapy services
- Has **“State of Art” Medical Technology**

HOSPITAL DEPARTMENTS

- Neuro Onco Surgery
- Ortho Oncology
- Plastic & Reconstructive Surgery
- Medical Oncology
- Bone Marrow Transplant (BMT)

- Radiation Oncology
- Haematology-oncology
- Gynaecologic Oncology
- Gastroenterology
- Anaesthesiology

- Pathology Lab
- Microbiology
- Histopathology
- Blood Bank

- Radiology
- MRI
- PET
- ECG , ECHO

BED CHARGES WITH DOCTORS VISIT CHARGES:

Name Of Ward	Bed Charges	Medical ICU	Surgical ICU	Additional Charges	Dr. Visit Charge
Day Care (A/C)	800	–	–	–	–
Day Care Deluxe	1200	–	–	–	–
General Ward (A/C)	1600	5000	3000	–	550
AML ward	2000	5500	3000		600
Cubicle	2000	4500	4500	25%	650
Semi Deluxe	3000	6000	5500	35%	650
Deluxe	6000	6500	6500	50%	650
Super Deluxe	9000	6500	6500	100%	1000
- Additional 10% service charge is taken on total bill					

Academics:

- **Offers Post graduates courses** in discipline of Surgical Oncology, Medical Oncology, Radiation Oncology, Pathology, Palliative Medicine,
- **22 seats**
- DNB, NBEMS Diploma Course in Anesthesiology and Radio-diagnosis and & FNB course.

PATIENTS PROFILE:

- **Economic Status:**

- 60-70% of patients are lower middle class to upper middle class
- 15-20% BPL
- 10-15% upper middle class and elites
- 2-3% international medical tourists

- **Paying Mechanism:**

- 60%-70% cashless through government insurance schemes or private insurance schemes (partly cash is being paid by most of them for OPD & diagnostic services and for difference between approved insurance amounts and actual bill)
- 30%-40% cash patients

- **Drainage Area:**

- Jaipur & Entire Rajasthan
- Neighbouring part of MP, Gujarat. UP, etc.

- **Reason for Choosing BMCHRC:**

- Comprehensive Cancer services with all the services under one roof
- High quality doctors
- Cashless services: empanelled with all available government schemes like PMJAY, CGHS, ESIS,

- Multiple corporate tie up with public and private sector entities
- Branding since last 28 years
- Only large private hospital in North West India
- Relatively cheaper than hospitals in Delhi or Mumbai
- Good patient care with trained staff

➤ **RENOVA RHL CANCER CENTRE, JAIPUR**



Radiotherapy centre with PET in RHL Hospital is being managed by Renova Cancer Centre
Medical Oncology and surgical oncology are being referred to RHL.

➤ **FACILITY PLAN COMPARATIVE: JAIPUR & KOTA CANCER HOSPITALS**

Name of Hospital	Bhagwan Mahavir Cancer Hospital & RC Jaipur	RHL Renova Cancer Centre (Raj. Hospital) Jaipur	American Oncology Institute, Srij Hospital. Kota	Sudha Medical College Hospital Kota	Proposed Indorama Cancer Institute Kota
Year of Establishment	1992	2017		2023	2027
Revenue INR Crores Approx.	700	NA	NA	NA	
Total Manpower	1050				
Total Bed	300		200	350	100
Medical ICU	16		Y	10	Y
Surgical ICU	16		N	10	Y
Bone Marrow Transplant Unit	6		Y	N	Y
Isolation Rooms	4		N	2	Y
Chemotherapy Unit	22		4	6	Y
General Ward	72		Y	30	Y
Sharing Room	32		Y	Y	Y
Special Room	32		Y	Y	Y
Deluxe Room	6		Y	N	Y

Super Deluxe/Suites	6		Y	N	Y
Operation Theatres	Y		Y	1	Y
Out-Patient Department	22		3	2	Y
DIAGNOSTIC s					Y
Clinical Pathology	Y		Y	Y	Y
Haematology	Y		Y	Y	Y
Biochemistry	Y		Y	Y	Y
Microbiology	Y		Y	Y	Y
Histology	Y		Y	N	Y
Blood Bank	Y		Y	Y	Y
Imaging Department					
X-ray	2		Y	Y	Y
Sonography	3		Y	Y	Y
C.T. Scan	Y		Y	Y	Y
Mammography	Y		Y	N	Y
MRI	Y		N	Y	Y

PET Scan	Y	Y	Y	N	Y
Nuclear Medicine	Y		N	N	
Nuclear Scan	Y				N
Radio-Immuno-Assay	Y				N
Radiotherapy					
Linear Accelerator	3	1	Y	1	Y
Brachy Therapy	Y	1	Y	1	Y
ECG	Y		Y	Y	Y
Echocardiography	Y		Y	Y	Y
Pul. Function Test	Y		Y	Y	Y
Physio with Rehab	Y		Y	N	Y
Emergency	8		4	5	4
24 Hours Pharmacy	Y		N	Y	Y
Support Services					
Kitchen	Y		Y	Y	Y
Laundry	Y		Y	Y	Y
CSSD	Y		Y	Y	Y

PREVALENT SALARY STRUCTURE:

Medical Staff	Salary Structure
Consultant Doctors: General Disciplines	250,000-350,000
Onco-Surgeons	400,000 – 1,000,000
Oncologist	350,000-600,000
Radiotherapist	300,000 – 450,000
Physicist	125,000 – 150,000
Centre Head	200,000 - 300,000
Staff Nurses	14.000 - 27,000
Supervisors / In-charge Nurses	35,000 – 55,000
Resident Doctor - MBBS	50,000-70,000
Radiotherapy Technicians	35,000-60,000
General Technicians	18000 - 35000
Admin Staff	
Manager	45,000-70,000
Assistant Manager	35,000 -45,000
General Supervisors	25,000-35,000
Clerks / Secretaries	15,000-25,000

➤ **PRODUCTIVITY INDICATORS: Comparative between Jaipur & Kota**

Productivity / day	BMCHRC	SUDHA MEDICAL COLLEGE – ONCOLOGY DEPARTMENT
OPD / day	1000-1200	15-20
Admission	35-40	6-7
Occupancy %	70% - 75%	50%-55%
Pathology Lab Samples / Day	400 - 500	NA
X-ray	50-60	30-40
USG	8-10	6-8
CT Scan	30	15-20
MRI	16	3-4
PET Scan	12	na
Chemotherapy	25-30	5-6
Radiotherapy	180-200	80
Major Surgery	18	15 (per mth)

➤ **TARIFF STRUCTURE:**

For Private Insurance, Corporate patients or cash patients

	BMCHRC
OPD	700 - 900
Pathology Lab	350
X-ray	350
USG	900 - 1400
CT Scan	2500 - 5000
MRI	4500 - 8000
PET Scan	12000 -18000
Chemotherapy	50,000 – 100,000
Radiotherapy per sitting	7000-9000
Major Surgery	40,000 -90,000
Health Checkup (Preventive cancer screening)	3000 – 12,000

CONSULTANTS & STAKE HOLDERS SURVEY:

- Survey was conducted as informal interviews visiting them at their clinics, hospitals, etc
- General Physicians in Baran & Bundi
- **Survey Findings:**
 - ✓ **1. Patients seen per day - OPD**
 - ✓ Good OPD numbers 10-15, even crossing 20 mark in case of few oncosurgeons.
 - ✓ Lowest 6 of an oncologist
 - ✓ Most of the doctors are having their own clinic cum hospital. Most of these consultants are attached to 2 to 3 hospital and there they see patients that are referred or through appointment
 - 1. Average number of Admissions & Hospital Preference**
 - ✓ 1-2 per day
 - ✓ Cash patients are admitted at their own hospitals and cashless patients are mainly admitted at **Sudha Hospital** or Sudha Medical College, some to Sriji, Ethos, etc
 - 2. Referral to Consultants:**
 - ✓ 50% of patients are referred by local consultants and by consultants from Baran, Bundi, etc

3. Willingness to pay for good quality healthcare

- ✓ Willing to pay only for Radiotherapy services
- ✓ For Surgical cases they tend to go to Jaipur to avail high quality services

4. Criteria for selection of Hospitals:

- Doctor's advice
- Empanelment with govt. and private insurance schemes
- Availability of all services under one roof
- Brand Name of hospital

5. Socio Economic Status of Patients:

- ✓ 60-70% of the patient belongs to BPL
- ✓ 20-30% of lower middle class
- ✓ 10%-50 % of middle class.
- ✓ <5 higher middle class and Elite class, mainly as first visit

6. Scope of Cancer Hospital:

- ✓ All the consultants opined for need of a high end cancer hospital providing cost effective quality care so that patients need not to travel to Jaipur or Delhi.

✓ **Patient Volumes:** *All the doctors were positive based on patient volume; Reason: no corporate level cancer hospitals in entire Hadoti Region with 7.0 population*

- **Increasing incidence of cancer**

- *Large referral base from secondary and extended secondary service area*

✓ **Commercial Viability:**

Viable due to high patient volumes and multiple government insurance scheme supporting cancer patients

7. Deficiency in Existing Healthcare:

- ✓ High Quality Nursing Care

- ✓ Quality Critical Care Services

8. Success Criteria:

- All the services under one roof
- Quality Healthcare Infrastructure & Medical Technology
- Empanelled for government schemes
- Corporate tieup with all local industries
- High Quality Service Delivery
- Affordable cost
- Good Consultants

RESEARCH ANALYSIS OUTCOME - PERTINENT OBSERVATIONS:

- All the areas in and around Kota are developing fast
- Disorganised primitive cancer services except for radiotherapy
- No high quality large cancer hospital in entire region.
- Current hospitals services are not sufficient for large population and they do not provide quality care
- Highly cost sensitive population however government schemes are there and private insurance penetration is increasing fast
- Full time availability of specialists
- Even BPL population will prefer good private hospitals empanelled with PMJAY Scheme i.e Ayushyaman Bharat Scheme, RGHS, etc.
- Alarming rise in cancer patients more so in Rajasthan
- Consultants are highly positive due to high volume of patients who are mostly going to Jaipur provided a corporate type dedicated cancer hospital is developed

➤ **MEDICAL INSURANCE SCENARIO / PAYMENT MECHANISM**

- Almost 60% of the population is covered under government or private health insurance
- PMJAY empanel private hospitals as well

GOVERNMENT INSURANCE SCHEMES:

1. Ayushman Bharat - Pradhan Mantri Jan Arogya Yojana (AB-PMJAY)

- Health cover of up to Rs. 5,00,000 per family per year to entire BPL population with 200 cancer treatment packages
- Additional cover of 5.0 for senior citizens from all economic strata
- Cashless access to hospitalization services for all cancer treatment modalities that is chemotherapy, radiotherapy and surgeries
- No cap on family size, age or gender
- Benefits across the country in all empanelled hospitals
- PM-JAY covers cost of hospitalization, treatment, up to 3 days of pre-hospitalization and 15 days of post hospitalization follow-up care
- Availability of affordable oncology medicines at Jan Aushadi and AMRIT pharmacies.

Coverage: All poor and vulnerable families are covered as per specific deprivation criteria identified under the Socio-Economic Caste Census (SECC), 2011.

2. Central Government Health Scheme (CGHS)

- ✓ Central Government Health Scheme (CGHS) is a health scheme for serving/retired Central Government employees, Member of Parliament (present and past) and freedom fighters with families.
- ✓ All the categories of people covered under the scheme need to subscribe for CGHS membership: The subscription rates varies depending on the Levels in the Pay Matrix.

3. Employees' State Insurance Scheme (ESIS)

- ✓ ESIS (Employees' State Insurance Scheme) covers cancer treatment. It provides comprehensive medical benefits including chemotherapy, radiation therapy, and surgery at empanelled hospitals. The scheme includes coverage for major cancers such as liver, pancreas, oral, breast, colorectal, and gynecological cancers. ESIS also offers extended sickness benefits for long-term and malignant diseases, including cancer, with wage support during treatment.

- ✓ ESIS is a contributory scheme on the patterns of social health insurance. The existing wage limit for coverage under the Act is Rs. 21,000/- per month (w.e.f. 01/01/2017).
- ✓ There is no ceiling on expenditure on the treatment of an insured person or his family member.

4. **Mukhyamantri Ayushman Arogya Yojana:**

- ✓ Offers cashless treatment up to ₹25 lakh per family
- ✓ Covers all type of cancer treatment
 - ✓ This new scheme effectively replaces BSBY and offers significantly more coverage for families in Rajasthan

Coverage:

- Entire BPL population and government employees, premium paid by government
- Rest of entire population after paying premium of Rs. 850 per year

➤ **Chief Minister Relief Fund**

- Additional financial support for cancer patients needing treatment, medicines, and travel assistance is available through the Chief Minister's Relief Fund.

➤ **Recent Initiatives in Budget**

- The 2025 Union Health Budget introduced plans for **customs duty exemptions on life-saving cancer drugs.**

➤ **CRITICAL OVERVIEW & SUCCESS CRITERIA**

Reason for Developing Hospital in Kota i.e. Why Kota?

1. Primary reason for hospital development is **deficiency of cancer hospital** in entire primary and secondary service area witnessing high incidence of cancer patients
2. Kota is the **largest and fast developing city in the area**

➤ **Cost of Land:**

In major cities, cost of land is very high and it increases the risk associated with big projects, cost is relatively much low in the Kota.

➤ **Paying Capacity:**

- 50% of the population in the region is lower middle & middle income class.
- Area is witnessing increasing population of upper middle class & Elites
- Government insurance schemes covering cancer treatment in private hospitals

➤ **Availability of Consultants, Nurses & Paramedics:**

- ✓ Kota is already having about 10 cancer specialists, all types of general speciality and

superspeciality consultants.

✓ Due to an old government medical college, an number of cancer specialists from Kota are serving in cancer hospitals across India. Already about 10 cancer specialist are available in Kota

✓ Similarly Nurses & Paramedics are also available in the city.

➤ **Unethical Practice:**

It is relatively low in the region. If the proposed hospital is ethical and will try to connect directly with patients, it will be able to establish faster. It will also ensure support from good local doctors for high quality services

➤ **Patients going Outside Kota**

Most of patient for high quality care & super speciality services go to hospitals in Jaipur or Delhi city.

➤ **Financial Viability:**

From start of hospital, 5 established Consulting Cancer Specialist who are already part owners of 5 different small cancer hospitals, will be available to ensure to give head start

➤ **Productivity:**

Will be good due deficiency of cancer hospital in entire region covering population of >25 Million

High volume of other superspeciality services

Insurance coverage for about 60% of population

➤ **Residential Accommodation & Other amenities for Relatives:**

Availability of low cost high end residential accommodation in the service area to attract doctors.

➤ **RISK FACTORS:**

Cost of Project:

Project cost including land cost leads to major financial burden. Controlling cost in building infrastructure with optimum infusion in Medical Technology and quality manpower will pay dividends for early viability

Delay in Completion of Project:

There is no other private hospital project of corporate nature coming up in the region, any delay may lead to competition and first mover advantage

Delay in Arranging Requisite Finances:

During project execution and during hospital operations

Doctors Appointment:

Big Brand Doctors in initial period may not be ready to join the hospital but with good PR and marketing exercise while the project is under development will help in attracting them

Staff Appointment:

Appointment of experienced manpower with knowledge of locally available insurance schemes in front office management, billing, etc. Similarly experienced staff for purchase, HR, Speciality technicians, critical care nurses, etc will be the key to build confidence in the market.

Quality Healthcare Delivery:

Perception of high quality affordable service from inception helps in improving patient flow which further helps in fast operational breakeven. Ensuring high quality services from day one is a huge task followed by challenge of maintaining service quality.

➤ **SUCCESS FACTORS**

Project Execution

Project execution is the primary success factor of a project. It is the foundation of the project on which future success is dependent. It has following main components:

Project Planning

Well thought out project planning ensures success of the project in terms of:

- Facility Plan & its phasing
- Master Plan for building expansion
- Functionally efficient design with optimum construction
- Financial Plan
- Management Plan

Project Implementation

It encompasses development of project in terms of minimum time, cost and high quality

Level of Medical Technology

Technological edge will always give good results in long run. Quality and level of medical technology is important for right diagnosis and treatment outcome. But again the cost

benefit advantage in terms of usage and service charges has to be kept in mind before buying medical technology.

Experience of Managing Hospital:

All the three main promoters are directors with various hospitals and have experience of managing middle level hospitals.

Quality & Sufficiency of Manpower

It is the most important factor in hospital industry being a high skill service industry dealing with life and death of patients. Hiring high quality manpower and its deployment scheduling after doing detailed planning with job profile is of paramount importance in success of hospital project

Project Commissioning: System Oriented Management

Developing good protocol and policies I,e SOP's with their implementation will ensure development of high quality system based organization

Finance Management:

Most of the project fails due to bad financial planning i.e either cost overrun or delay in payments

Professional & Ethical Management:

Quality of Services i.e. Patient Satisfaction

Patient satisfaction is dependent on the quality of medical services, hospitality services and cost i.e value for money proposition. If these 3 are managed well and innovate taking continuous feedback, patient brings patients i.e market your services.

Success Criteria Mentioned by Doctors:

- All the services under one roof
- Quality Healthcare Infrastructure & Medical Technology
- Empanelled for government schemes
- Corporate tieup with all local industries
- High Quality Service Delivery
- Affordable cost
- Good Consultants

All the studied facts substantiate need for large corporate cancer hospital in the area, hence the proposal to develop a 100 bed World Class hospital.

PROJECT CONCEPTUALIZATION WITH FACILITY PLAN

✓ Objectives of the Project:

- To fulfil mission & vision of the promoters
- Make it a technically and financial sustainable project

A “Centre of Excellence” that provides Holistic Cancer Healthcare at affordable cost to become the preferred choice of Hospital in the primary and secondary service area.

Project has to be conceptualized keeping these objectives in mind.

✓ Project Concept:

The proposed project consists of:

- A 100 bed Cancer Hospital with “state of art building & medical technology”

✓ USP's of the Project:

- High End Corporate Hospital with elegant interiors in a composed building i.e. low average area per bed however no compromise in critical areas and circulation within the hospital
- Started by a HNI philanthropist family to give back to the society
- Provides complete spectrum of cancer services to prevention, cure and rehabilitation

➤ World class services still affordable to local community

✓ **Major Assumptions based on which facilities are being planned:**

Cancer Hospital:

- Approximately 7.0 million population not having quality cancer hospital services hence a hospital providing high end diagnostic services, medical oncology, surgical oncology Radiotherapy, etc. with high quality services are the need of the region
- High Population with high cancer incidence rates in need of quality services at affordable cost

Facility Plan:

- Strategic development of a facility plan is the key for success of the project.
- Important Observations to be considered for this development:-

✓ **Clients Objective:**

A financially self-sustainable and affordable World Class Cancer Hospital for “one & all”

➤ PROJECT CONCEPT

Location:

- ✓ Strategic
- ✓ Swami Vivekanand Nagar
Kota

Plot Area: 15,000 Sq. Meters

Shape: Rectangle, good for
hospital development

Proposed Hospital Size:

100 Bed

Type:

Cancer Hospital

Company Structure:

Section 8 Company – Not for
profit

Promoters: Indorama

Institute of Medical Science, Gurgaon



➤ **MEDICAL SERVICES PLAN**

	Number of Bed	100
		Specialities
1	Pathology with Cancer Markers	Medicine
2	Histopathology	Intensive Care
3	USG	Hemato-oncology & BMT
4	X-ray / CT Scan / MRI / Mammography / BMD	Medical Oncology
5	Linear Accelerator	Surgical Oncology
6	Brachy Therapy	Neuro-oncology
7	PET Scan	Ortho-oncology
8	Bone Marrow Transplant Unit	Palliative and Pain Management
9	Intensive Care	Gynaec-oncology
10	Day Care / Chemotherapy	Anaesthesiology
11	Operation Theatres	Plastic & Reconstructive Surgery
12	ECG, ECHO	Gastroenterology
	Emergency Department	

➤ **FACILITY PLAN**

BED	100
General Ward	20
Twin Sharing Room	20
Single Room	16
Deluxe Room	2
Suite Room	2
Daycare	8
Chemotherapy Unit	6
Isolation Bed	2
IMCU	10
SICU	10
BMT	4
Service Bed	8
Emergency	4
Preop / Post Op	4

Hospital Size	100 BED
Out-Patient Department	
Reception / Registration	
Consulting Rooms	12
Minor Procedure Rooms	1
Endoscopy Suites	1
Doctors Lounge	1
Staff Rooms	1
Health Check up	1
24 Hours Pharmacy	
DIAGNOSTIC DEPARTMENTS	
Imaging Department	
Digital X-rays	1
Mobile X-rays	1
Ultra Sonography	1
C.T. Scan	1
MRI	1
Digital Mammography	1
BMD	1
PET	

RADIOTHERAPY	
Linear Accelerator	1
Brachy Therapy	1
Stereostic Fram	1
ECG	1
Echocardiography	1
Pathology Lab	
Operation Theatre Complex	4
Physiotherapy Department with Rehab	
Support Services	
CSSD	1
Kitchen	1
Housekeeping Department	1
Amenities	
Coffee Shop	1
Bank / ATM	1
Staff & Relatives Accommodation	

➤ **THRUST AREAS:**

“Focused approach always lead to early success”

1 Chemotherapy:

- Cost effective chemotherapy with discounted cost of drugs
- Patients from Hadoti region undergoing surgeries in Jaipur or other locations can be targeted for chemotherapy.

2 Radiotherapy:

- Low cost radiotherapy with high end technology could be used to market radiotherapy
- Patients from Hadoti region undergoing surgeries in Jaipur or other locations can be targeted for radiotherapy.

3 Bone Marrow Transplants:

India having high population of children with high incidence of leukaemia. A dedicated consultant with training from abroad will give good results which will boost success

4 Breast Cancer & Cervical Cancer:

High incidence hence the department should be developed fast with good doctors

5 Head & Neck Cancer:

High incidence of oral cavity and lip cancers augurs well for the success of the department

6 Preventive Oncology (Cancer Screening Health Checkups)

With increasing awareness about cancers and with further educations and awareness programs people could be motivated to go for preventive screening.

7 Telemedicine

While no panacea, telemedicine has the potential to liberate services from traditional settings, transforming small and otherwise vulnerable facilities into cost-effective and clinically safe components within a health care system, effectively decentralizing expertise.

Telemedicine would seem to have the potential to address many of the health care current challenges:

- Increased consumer expectations and in particular, reluctance to travel
- Cost effective, saves patients from hospital visits
- This will also be in line with the government aided projects whereby distant secondary level hospitals in the state can be connected to the hospital for a teleconsultations to cancer patients

➤ **BENEFITS OF DOING PROJECT UNDER TRUST**

- **Tax Exemptions:** Tax exemption under Section 12A/12AB of the Income Tax Act, 1961, which provides a complete exemption from income tax on all income used for charitable purposes.
- **Tax-Deductible Donations (Section 80G):** Section 80G, allow donors to claim tax deductions on their contributions. Important for raising funds.
- **Tax Exemptions to Research Centers (registered with Dept. of Industrial Scientific Research (DISR):** Section 35 for research funding, contributions act. Up to the extent of 150%
- **Access to Funding : from** private foundations, CSR Funds and foreign contributions (with FCRA registration) that are often unavailable to for-profit entities.
- **Enhanced Public Trust and Credibility :** being a non-profit organization
- **Asset Protection:** A trust structure provides protection for the trustees' personal assets
- **Perpetual Existence:** trust as a legal entity can continue its operations regardless of changes in trustees, ensuring long-term sustainability and continuity.

➤ **LEGAL & STATUTORY REQUIREMENTS:**

1	Nursing Home / Health License	Public Health Department, Local Municipal Office	License
2	License for Retail Pharmacy	District Drugs Controller, FDA	License
3	License for Wholesale	District Drugs Controller, FDA	License
4	License for Alcohol - Rectified Spirit II	Excise Department	License
5	License for Spirit - Denatured Spirit – IV	Excise Department	License
6	Fentanyl & Pethidine Drug Transportation License	FDA	License
7	License to sell, stock or exhibit or offer for distribute by retail the drug specified in Schedules C, C (1) and X/ Homeopathic medicines	FDA	License
8	NDPS-I: License for the manufacture, possession and sale, otherwise than on prescription of manufactured drugs by dealers	FDA	License
9	NDPS-II :- License for the possession and sale, on prescription of manufactured drugs by chemist and medical practitioners	FDA	License
10	Blood Bank/ Blood Storage Center	FDA	License

11	Food Inspector License for Eating House & storage of kitchen items	FDA	License
12	Canteen PFAA Registration/ License & Storage of Perishable Food Items	FDA	License
13	CT Scan, X-Ray, etc.	AERB	Approval
14	PET, Nuclear Medicine & Radiotherapy	AERB	Approval
15	Registration under Shops & Commercial Establishment Act for Hospital	Local Municipal Office	Registration
16	USG/ 2D Echo (PNDD)	Local Municipal Office	Registration
17	Nursing Council Registration	Nursing Council	Registration
18	Authorization for Bio-Medical Waste Disposal	Pollution Control Board	Registration
19	Authorization for Bio-Medical Waste Generation	Pollution Control Board	Registration
20	DG Set	Pollution Control Board	Registration
21	STP	Pollution Control Board	Registration
22	Guest House / Hospice	Registrar, Department of Registration & Stamps	Registration
23	Ambulance Vehicle	RTO	Registration

24	Vehicle Ambulance	RTO	Registration
25	Registration of Staff	Employment Exchange	Registration
26	Employees Provident Fund Registration	PF Commission	Registration
27	Department of Health	State Government	Registration
28	Organ Retrieval	ZTCC	Registration
29	Organ Transplantation	ZTCC	Registration
30	Occupancy/ Partial Occupancy Certificate	Local Municipal Office	Certification
31	Contract Labour Certification - Principal (out-sourced)	Labour Commissioners Office	Certification
32	Elevators & Escalators	GOM, Industries, Energy and Labour Dept.	Certification
33	ESIC Exemption	ESIC Office	Certification
34	Liquid Medical Oxygen	CFO	Certification
35	Fire NOC	CFO	Certification
36	Boiler	CFO	Certification

37	Neon Light/ Sign Board	Local Municipal Office	Certification
38	Vehicle Road Tax	Municipal Corporation	Statutory
39	Building & Equipment Insurance	Private	Statutory
40	Vehicle Insurance	Private	Statutory
41	Management of HIV Cases	NACO (MCI Directive)	Notifiable
42	Medical Tourism	Police, Consulate	Notifiable
43	GST for Pharmacy, Property Tax, PAN, 35 AC (Research Center), 80G (Charity), Professional Tax	Respective Government Department	Statutory

➤ Source: Hospihealth Database

GOVERNMENT INSURANCE SCHEMES & CORPORATE TIE UPS

Operating in the region for Hospital Services

• RGHS • CGHS • Maa Yojna • ESIC • ECHS • NPCIL (RAPP) • Railway • CFCL • NTPC	• BSNL • ACC Cement • Airport Authority of India • GAIL • SAIL • PowerGrid • Bank of Baroda • Punjab National Bank • Rajasthan Housing Board	• SBI • Corporative Bank Society • DCM Shriram Polyclinic • J.V.V.N.L • K.T.P.S. • Krishi Vipdan Board • RBSK • RIICO • Tilhan Sangh, Kota
Insurance companies & TPA's working in the region		
• ACKO General Insurance • Aditya Birla Health Insurance • Bajaj Allianz General Insurance • Care Health Insurance • Cholamandalam MS General Insurance	• ICICI Lombard General Insurance • IFFCO Tokio General Insurance • Kotak Mahindra General Insurance • Liberty General Insurance • Magma HDI General Insurance	• SBI General Insurance • Star Health & Allied Insurance • Tata AIG General Insurance • The New India Assurance Co. • The Oriental Insurance Company

- Future Generali Insurance
- Go Digit General Insurance Ltd
- HDFC ERGO General Insurance
- Reliance General Insurance

- National Insurance Company
- Navi General Insurance
- Niva Bupa Health Insurance
- Royal Sundaram Gen Insurance

- United India Insurance Co.
- Universal Sompo General Insurance
- Zuno General Insurance

Dear All,

Please call for any clarification.

Regards,

Dr. Vinay Kothari

PART 2:

FINANCIAL FEASIBILITY REPORT

Objectives of Financial Feasibility are to:

- Arrive at **project cost** for the proposed Facility-mix as per the envisaged budget
- Commercial **viability** and **sustainability** of the project based on the given means of finance

FINANCIAL PROJECTIONS

The most important part of the project report is the financial projects and the financial analysis. It requires a great deal of experience to create this report since there could be a number of permutations and combination of services, their capital cost, income & expenses. These projections are an effort towards the objective of attaining a near accurate picture. The whole report is backed by a scientifically performed market survey which gives input in terms of project size, facility plan, building & technology required, manpower, salary structure, tariff structure, productivity pattern, etc. to compute project cost and income & expenditure estimates. The most crucial part of the report is the assumptions considered based on the market survey, market potential as well as the knowledge and experience of the consultants.

It is only on the basis of this report, promoters can take appropriate investment decisions for the project.

- **Methodology**
- The data for the projects are collected through an exhaustive market survey.
- The same is then analysed to develop the facility plan, level of technology required, forecasting income and expenditure based on assumptions.

- All projections are based on the current rates prevailing in the geographic service market.
- The rates are considered relatively on the lower side to beat the market competition and optimum utilization of services.

- **Assumptions:**

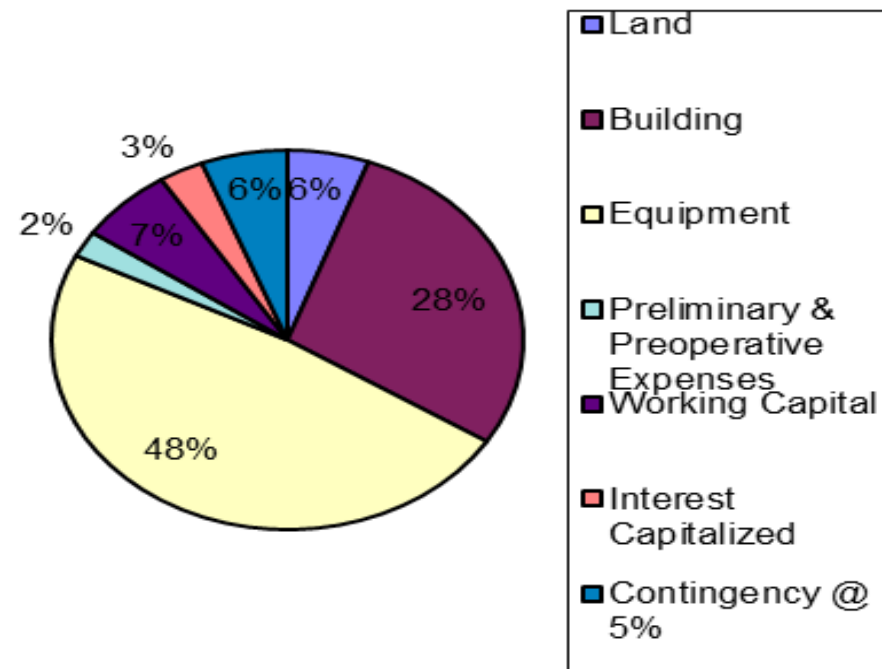
Based on the above study, the important broad assumptions are:

- Facilities are proposed keeping in mind the International Standards and the requirement of the service area
- Revenue projections are based on the rates prevalent in similar hospitals, productivity, level of technology used, aesthetics and medical care quality.
- Expenditure assumptions are based on comparisons of current working patterns in other hospitals, technology, human resource utilization, and expected productivity.
- For computation of income projections, the tariff structure and average productivities of services hospitals in the service area were studied in detail.
- To arrive at the prices of the various facilities proposed, tariff structures comparison of the services of hospital along with the quality of similar services in the service area was compared for each procedure.
- The income and expenditure have been distributed, based on the number of occupied beds and outpatient productivity over the years.
- An inflation of 7% per annum is considered in both income and expenditure estimates from third year onwards as in the initial phase when hospital is taking off, it is not advisable to revise the rates

1. COST OF PROJECT:

The total project of the hospital is phased out over three years inclusive of building & technology so as to cut down on initial investment and also to take care of gestation period.

COST OF PROJECT: PERCENTAGE DISTRIBUTION



COST OF PROJECT AND MEANS OF FINANCE				
		Rs. Lakhs		
		Phase 1	Phase 2 (3rd Years)	
Land		500		
Building Development				
Main Building		2,539	748	
Sub Total		2,539	748	
Medical Equipment		4,798	405	
Non-medical Equipment + Vehicles		124	58	
Furniture		82	71	
Sub Total		5,004	533	
Total		7,543.20	1,280	
Total EPC				
Preliminary & Preoperative Expenses @ 1%		75.43		
Working Capital (3 Mth)		538		
Contingency @		7%	528	
IDC		274.33		
Sub Total		9,458.75	1280.37	
Total				
GRAND TOTAL		10,739		

- **Land:** The land will be used to create Cancer Hospital and related ancillary and support facilities.

LAND & BUILDING DEVELOPMENT

INR

Land Area Sq. Meter	15,000
Cost INR Crores	500

Total Bed	100
	Sq. Ft.
Area Per Bed	700
Bed Area	70,000
Basement	20,000
Total Area	90,000

		Phase 1	Phase 2		
Total Construction Area Sq. M					
Total Construction Area Sq. Ft		90,000	0		
Cost of Construction		Phase 1		Phase 2	
	Rate	Qty.	Rs.	Qty.	Rs.
Civil	1,400	90,000	126,000,000	0	0
Interiors	750	54,000	40,500,000	36,000	27,000,000
Plumbing / Fire Fighting	350	54,000	18,900,000	36,000	12,600,000
Electrical	350	54,000	18,900,000	36,000	12,600,000
HVAC	350	54,000	18,900,000	36,000	12,600,000
Elevators	32,00,000	3	9,600,000	2	6,400,000
Landscape & Site Development	100	90,000	9,000,000		0
Sub Total			241,800,000		71,200,000
Consultants/ Architects's Fees	0		12,090,000		3,560,000
TOTAL			253,890,000		74,760,000

➤ **Building Development**

- The total building project is about 85,000 sq. ft.
 - Hospital Building: 65,000 sq. ft.
 - Basement of 20,000 Sq. ft.
 - The Hospital building complex will have 100 bed inpatient facilities, Diagnostic services and other support services.
 - Basement will be used for Creating Radiotherapy in 8,000 sq. ft.
 - Rest of basement will be used for, Engineering services, Support services and Parking.
- The whole project will be commissioned in 2 phases.
- Phase 1 will have entire basement and entire building structure.
- 60% of the area inclusive of basement will be made ready by doing interiors and MEP services to house Radiotherapy, diagnostic services along with 50 inpatient bed.
- Rest 40% of the area will be made ready in third year to house another 50 bed.
- The cost of construction is considered based on the prevalent market rates and past experience of the consultants.
- The cost include building with bunkers, interiors, services like electrical, air-conditioning, plumbing, elevators, site development, fixtures, water tanks, painting etc.
- Suitable mark-up for contingency and architect's fees is also considered.

➤ **Equipment**

Hospitals are highly capital and technology intensive. With this backdrop it is very important to invest wisely in medical & non-medical equipment without compromising on quality. Utmost care has been taken to select appropriate equipment, considering capacity utilization, availability of professionals to operate, demand, and need.

MEDICAL EQUIPMENT					
<u>Operating Theatres</u>	Unit Rate	Phase 1		Phase 2	
		Qty	Value	Qty	Value
Hydraulic Operating Tables (Multipurpose)	14,00,000	2	28,00,000	2	28,00,000
Ceiling Mounted lights[Imported]	8,50,000	2	17,00,000	2	17,00,000
Anaesthesia machine	14,00,000	1	14,00,000	1	14,00,000
Advanced Boyles	3,00,000	1	3,00,000	1	3,00,000
Multiparameter Monitor(Cardiac)	3,50,000	2	7,00,000	2	7,00,000
Patient Warming Blanket	1,50,000	1	1,50,000	1	1,50,000
Electrocautery (Imported)	8,50,000	2	17,00,000	2	17,00,000
Harmonic Scalpel	35,00,000	1	35,00,000		0
Syringe / Infusion Pump	45,000	8	3,60,000	8	3,60,000
Suction Machines	20,000	2	40,000	2	40,000
Defibrillator / Monitor / Recorder	3,75,000	1	3,75,000	1	3,75,000
Mobile Light	1,50,000	2	3,00,000	2	3,00,000
Flash Steriliser (Instrument)	2,25,000	1	2,25,000		0
Laposcopic Surgery Set	52,00,000	1	52,00,000		0
Surgical Instruments	5,00,000	2	10,00,000	2	10,00,000
Integrated Modular OT	30,00,000	2	60,00,000	2	60,00,000
Miscellaneous		5%	9,87,500		5,41,250
Total			2,67,37,500		1,73,66,250

ICU's (10 Bed Each)

Bedside Monitors (multiparameter)	2,50,000	10	25,00,000	10	25,00,000
Pacemaker	1,50,000	1	1,50,000	1	1,50,000
Central Station	7,50,000	1	7,50,000	1	7,50,000
Syringe/ Infusion Pumps	35,000	20	7,00,000	20	7,00,000
Pulse Oximeter with Recorder	1,20,000	2	2,40,000	2	2,40,000
Defibrillator/ Monitor / Recorder	3,75,000	1	3,75,000	1	3,75,000
Ventilator	12,00,000	2	24,00,000	2	24,00,000
BIPAP	4,00,000	2	8,00,000	2	8,00,000
Nebulizer	10,000	2	20,000	2	20,000
ECG Machines (12 channel)	2,25,000	1	2,25,000	1	2,25,000
Suction Machines	20,000	2	40,000	2	40,000
Miscellaneous		5%	4,10,000		4,10,000
Total			86,10,000		86,10,000

Radiology

MRI (1.5 tesla)	45,000,000	1	45,000,000		0
CT Scan 128 Slice	35,000,000	1	35,000,000	0	
USG with Color Doppler	4,800,000	1	4,800,000		0
Digital Mammography	7,000,000	1	7,000,000		4,200,000
BMD	750,000	1	750,000		0
X-ray	900,000	1	900,000		0
Portable X Ray 60 mA	225,000	2	450,000	1	225,000
CR System	2,300,000	1	2,300,000	1	2,300,000
X-ray viewer [8 film panel]	65,000	2	130,000	1	65,000
Miscellaneous		5%	4,816,500		339,500
Total			101,146,500		7,129,500

RADIOTHERAPY (Out Sourced)

LINAC	180,000,000	1	180,000,000		
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CT Simulator	18,000,000	1	18,000,000		
Brachy	25,000,000	1	25,000,000		
PET Scan	75,000,000	1	75,000,000		
Mis:		5%	11,150,000		0
			309,150,000		0

Cardiology Equipment

IABP	3,200,000	1	3,200,000		
ECG	220,000	1	220,000		0
Miscellaneous		5%	171,000		0
Total			3,591,000		0

Endoscopy Dept.

Endoscopy Set	4,500,000	1	4,500,000	1	2,500,000
Miscellaneous		5%	225,000	5%	125,000
Total			4,725,000		2,625,000

Laboratory

Automatic Clinical Analyser	2,700,000	1	2,700,000		0
Cell Counter 5 part	1,500,000	1	1,500,000		0
Blood Gas Analyzer	850,000	1	850,000		0
Electrolyte Analyser	550,000	1	550,000		0
Binocular Microscope	200,000	1	200,000		0
Microscope	60,000	1	60,000		0
Automated Blood Culture System	3,600,000	0	0	1	3,600,000
Elisa Reader & Washer (Microwell	550,000	1	550,000		0
Incubator & hot air oven	50,000	1	50,000		0
Centrifuge machine	10,000	1	10,000		0
VDRL rotator	5,000	1	5,000		0
Flame photometer	30,000	1	30,000		0
Lab autoclave	50,000	1	50,000		0
Blood Bank	4,500,000	1	4,500,000		0
Miscellaneous		5%	552,750		180,000

Total			11,607,750		3,780,000
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Physiotherapy

Short Wave diathermy (500 watt)	250,000	1	250,000		0
Pulsed Short Wave diathermy (500 watt)	375,000	1	375,000		0
SWD (250 watts)	150,000	1	150,000		0
Ultrasound therapy unit (Dula frequency)	65,000	1	65,000		0
Muscle stimulator (Multiparameter)	45,000	1	45,000		0
TENS (Dual)	4,500	1	4,500		0
Interferential Current Therapy unit with vacuum (Dual)	225,000	1	225,000		0
Nebulizer	25,000	1	25,000		0
Continuous passive motion unit	275,000	1	275,000		0
EMG biofeedback	75,000	1	75,000		0
Wax bath	25,000	1	25,000		0
Miscellaneous		5%	75,725		0
Total			1,590,225		0

Dental Dept.

Dental Chair with Xray	550,000	1	550,000		0
Instrument Set	100,000	1	100,000		0
Flash Steriliser (Instrument)	125,000	1	125,000		0
Miscellaneous		5%	38,750		0
Total			813,750		0

Dialysis

Dialysis Machine	950,000	2	1,900,000		0
Dialyser	300,000	1	300,000		0
Mixer	150,000	1	150,000		0
Miscellaneous		5%	117,500		0
Total			2,467,500		0

Emergency Department

ECG Machine (12 channel)	60,000	1	60,000		0
Multiparameter Monitor	150,000	4	600,000		0
Transport Ventilator	650,000	1	650,000		0
Defibrillator/ Monitor / Recorder	375,000	1	375,000		0
O.T.Table	300,000	1	300,000		0
OT light	180,000	1	180,000		0
Advanced Boyles	350,000	1	350,000		0
Nebulizer	20,000	1	20,000		0
Suction Machine	20,000	1	20,000		0
Miscellaneous		5%	127,750		0
Total			2,682,750		0

CSSD

Steam sterilizer	1,800,000	1	1,800,000		0
Standby Cylindrical Steriliser	250,000	1	250,000		0
ETO Steriliser	550,000	1	550,000		0
Flash Steriliser	125,000	1	125,000		0
Ultrasonic Instrument washer /cleaner	650,000	1	650,000		0
CSSD furniture & fixtures	900,000	1	900,000		0
Miscellaneous		5%	213,750		0
Total			4,488,750		0

DU Equipment

Bed Pan Washer Disinfector	200,000	6	1,200,000		0
		5%	60,000		0
Total			1,260,000		0

Nurse Call system

Ward and ICU	18,000	50	900,000	50	900,000
		5%	45,000		45,000
Total			945,000		945,000
Medical Equipment	Total		479,815,725		40,455,750

NON MEDICAL EQUIPMENT

Medical gases, compresor & piping			36,00,000		18,00,000
Pneumatic Shute Sytsem			0		
<u>General Administration,</u>					
<u>Computers & peripherals & Software</u>					
Computer lan Network & software	30,00,000	1	30,00,000		
Hardware	15,00,000	1	15,00,000	6,00,000	15,00,000
Communications	7,50,000	1	7,50,000		
Total			52,50,000		15,00,000

Laundry

Total	18,00,000	1	18,00,000		
Morgue	5,50,000	1	5,50,000		
<u>Kitchen Equipment etc</u>	12,00,000	1	12,00,000	1	12,00,000
Non-Medical Equipment Total			1,24,00,000	1	45,00,000

AUTOMOBILES

Ambulances (Cardiac)	27,00,000		0		0
General Ambulances	12,00,000		0	1	12,00,000
				1	0
Add : Contingency		5%	0		60,000
Total			0		12,60,000

➤ **Furniture:**

Furniture requirement is very high in hospitals as there are multiple sets required, whether it is patient rooms, patient transfer furniture, operation theatres, office furniture or storage racks. All the furniture have their own importance based on location. Patient's room furniture has to have high comfort levels and adjustments, similarly for operation theatres furniture has to be of stainless steel i.e. rust free and easy for cleaning. Office furniture has to be high on aesthetics, etc.

FURNITURE				
Patient related areas		72,29,000		70,61,000
Pharmacy		1,86,000		1,41,000
Central stores		2,20,000		1,10,000
Medical records		1,88,000		1,10,000
		78,23,000		74,22,000
ADD:				
Contingency	5%	3,91,150		3,71,100
Total		82,14,150		77,93,100

FURNITURE					
Details of Furniture					
		Phase 1		Phase 2	
Patient related areas	Unit Rate	Qty	Value		
Electronic Beds (Fully Automatic)	1,80,000	20	36,00,000	20	36,00,000
Electronic Beds (Semi Automatic)	1,10,000	10	11,00,000	10	11,00,000
Fowler Bed	45,000	20	9,00,000	20	9,00,000
Emergency Strecher Trolley	45,000	4	1,80,000	4	1,80,000
Strecher trolley	7,500	6	45,000	3	22,500
Wheel chairs	4,500	6	27,000	3	13,500
Instrument Trolley	6,500	6	39,000	3	19,500
Dressing trolley	6,500	6	39,000	3	19,500
Soiled Linen Trolley	5,500	1	5,500	1	5,500
Bedside lockers	4,500	58	2,61,000	50	2,25,000
Over bed Table	4,500	50	2,25,000	50	2,25,000
Waste receptacle	150	70	10,500	50	7,500
TV - 32 Inch	13,500	20	2,70,000	20	2,70,000
Fridge Nursing Station / ICU, etc	27,000	4	1,08,000	2	54,000
Fridge Patient Rooms, etc	7,500	10	75,000	10	75,000
OPD Tables	21,000	6	1,26,000	6	1,26,000
Doctors Chairs	3,000	6	18,000	6	18,000
OPD Instruments	2,00,000	1	2,00,000	1	2,00,000
Total			72,29,000		70,61,000
Pharmacy / Stores	Unit Rate	Qty	Value		77,93,100
Storage shelves with glass panels	30,000	2	60,000	2	60,000
Cupboards	18,000	2	36,000	2	36,000
Open Shelf	7,500	12	90,000	6	45,000
Total			1,86,000		1,41,000
Central stores					
Cupboards	22,000	2	44,000	1	22,000
Open Shelf	12,000	12	1,44,000	6	72,000
Filing cabinets	16,000	2	32,000	1	16,000
Total			2,20,000		1,10,000
Medical Records					0
SS filing cabinets (open type)	18,000	8	1,44,000	4	72,000
Cupboards	22,000	2	44,000	1	22,000
Total			1,88,000	112	94,000

➤ **Preliminary & Pre-Operative Expenses**

These have been taken into account for standard preliminary expenses, which are traditionally encountered prior to starting a project.

These would typically relate to:

- Project Consultant Costs
- Legal & Conveyance Fees
- Market Feasibility Reports

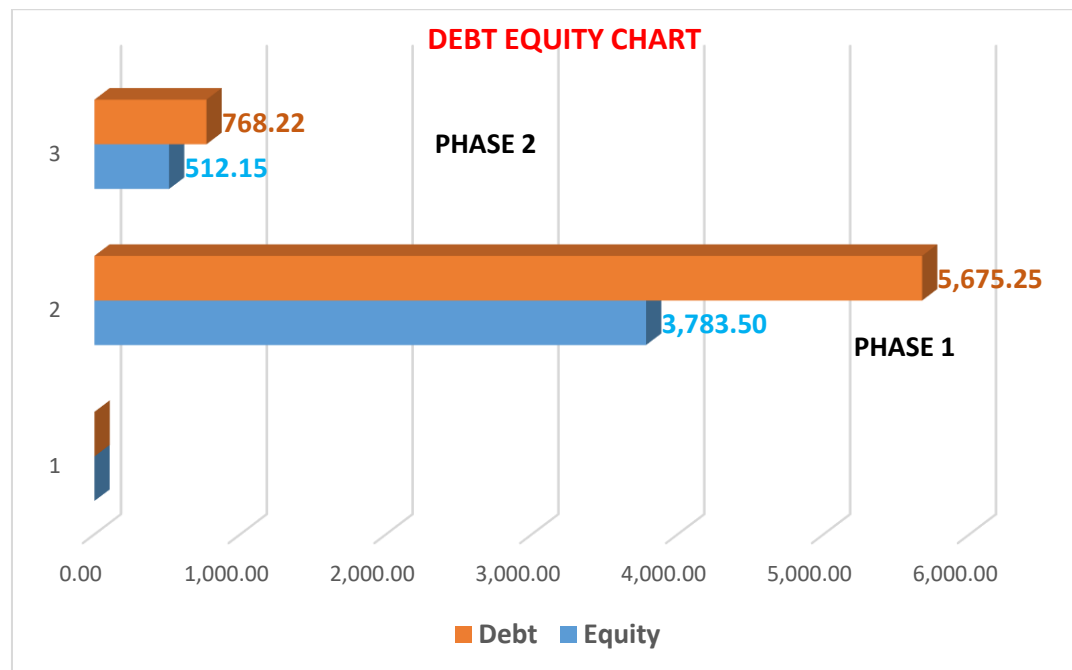
➤ **Working Capital**

It is essential to consider the requirement of working capital since hospital would initially require funds till the revenues begin to flow in. In this case, the working capital has been estimated at operating expenses for the first 3 months of operation. This has to be considered though the hospital would start receiving revenue from the first day itself.

2. Means of Finance

INR in Lakhs

MEANS OF FINANCE		Phase 1	Phase 2
Equity	40%	3,783	512
Debt	60%	5,675	768
Total	100%	9,459	1,280



INCOME ESTIMATES

Income shall be generated from the following two primary sources

INCOME ESTIMATES			
Phae 1 & Phase 2 are considered together starting operations at low occupancy levels .			
ROOM REVENUE			
Category	Beds	Rate	Total
Pay Bed	100		
Economy Ward	20	1,200	87,60,000
Twin Sharing Room	20	2,000	1,46,00,000
Single Room	16	4,000	2,33,60,000
Deluxe Room	2	6,000	43,80,000
Suite Room	2	9,000	65,70,000
Chemotherapy Chair	8	2,000	58,40,000
Daycare	6	2,000	43,80,000
Isolation Bed	2	12,000	87,60,000
IMCU	10	6,000	2,19,00,000
SICU	10	6,000	2,19,00,000
BMT (cost will be part of t	4		0
			12,04,50,000
Service Bed	8		
Emergency	4	0	0
Preop / Post Op	4		
TOTAL BED	108		
Charity @	10%		1,20,45,000
GRAND TOTAL			10,84,05,000

• Bed Revenue

• Department Revenue consisting of all diagnostic services, OPD revenue, OT revenue, ambulatory services and pharmacy with consumables.

The additional revenue has been considered from other sources like:

➤ Ambulance Services

➤ **Bed Revenue**

This has been calculated on the number of beds assumed to be occupied in a year, as a percentage for the beds available for that year. The number of beds increases from 1st year occupancy considered at 35% of the total beds to 90% considered in 7th year of operations. The total number of bed planned are 100 pay bed and 8 service bed.

	2,026	2,027	2,028	2,029	2,030	2,031	2,032	2,033	2,034	2,035
Number of beds	100	100	100	100	100	100	100	100	100	100
Capacity	30%	40%	50%	60%	70%	80%	90%	90%	90%	90%
Revenue	325	434	564	704	854	1,015	1,187	1,235	1,284	1,335
(Plus inflation @ 4% from 3rd year)										

➤ **DEPARTMENT REVENUE (Investigations & Treatment)**

This has been extrapolated in relation to the number of beds occupied from the various clinical departments of the hospital project in a given year and the total number of outpatients. Assumptions for revenue for certain diagnostic services are based on the market survey.

- Department revenue is a function of productivity of a service and the rate of that services
- Productivity of a service is a function of the percentage utilization of that service in a particular year and the total assumption considered for that service.

DEPARTMENT - Assumptions for Income

Assumptions	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10
Total no of days - Indoor	365									
Total no of days - OPD	300									
Average Length of stay	4.5									
Total number of beds	100	100	100	100	100	100	100	100	100	100
Occupancy	30%	40%	50%	60%	70%	80%	90%	90%	90%	90%
Beds Occupied	30	40	50	60	70	80	90	90	90	90
Beds Days	10,950	14,600	18,250	21,900	25,550	29,200	32,850	32,850	32,850	32,850
No of admissions	2,433	3,244	4,056	4,867	5,678	6,489	7,300	7,300	7,300	7,300

DEPARTMENT ASSUMPTION:

- This is the most important part of the feasibility as using this assumptions entire productivity calculations are done for each department at a given capacity utilization.
- There are 2 sets of assumption, one is for calculating capacity potential to calculate yearly revenue like OPD and IPD days in a year, admission capacity based on ALOS (average length of stay, capacity utilization envisaged based on healthcare industry experience, etc

DEPARTMENT - Assumptions for Income

Assumptions	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10
Total no of days - Indoor	365									
Total no of days - OPD	300									
Average Length of stay	4.5									
Total number of beds	100	100	100	100	100	100	100	100	100	100
Occupancy	30%	40%	50%	60%	70%	80%	90%	90%	90%	90%
Beds Occupied	30	40	50	60	70	80	90	90	90	90
Beds Days	10,950	14,600	18,250	21,900	25,550	29,200	32,850	32,850	32,850	32,850
No of admissions	2,433	3,244	4,056	4,867	5,678	6,489	7,300	7,300	7,300	7,300

- Second set of assumptions is sets of assumption are considered based on market research and benchmarking analysis along with industry standard for a particular department.

- These department wise assumptions gives total productivity assumption for a department in a particular year for OPD and IPD patients separately which are further added together to give total productivity of a department in a particular year

DEPARTMENTAL ASSUMPTIONS

PATHOLOGY	Indoor	OPD
Investigations per day per bed	3	1
Blood Components		
Per Day	24	
RADIOLOGY		
Avg. X Ray per admission	100%	10%
ULTRASONOGRAPHY		
per day	4	4
C.T. SCAN		
per day	8	16
MRI		
per day	8	8
BMD		
per day	2	2
Mammography		
per day	2	4
ECG		
	100%	10%
ECHOCARDIOGRAPHY		
per day	2	2
PHYSIOTHERAPY		
% of occupied beds	30%	12

OPD		
per day		240
Endoscopy		
per day	4	
PET Scan		
per day	6	6
Radiotherapy (Linec)		
per day		80
Brachytherapy		
per day		8
Minor OT / Biopsy per Day		
per day	4	4
Major Surgeries		
per day	2	
Supramajor Surgeries		
per day	6	
BMT		
Per day	0.030	
Chemotherapy		
Per Day		8
ADMISSIONS		
As per total admissions		
PHARMACY		
As per total bed strength % OPD	100%	
EXECUTIVE HEALTH CHECK		
per day		12

➤ **AVERAGE RATE: FOR A SERVICE**

AVERAGE RATE in INR		
IP Rate @ 25% higher than OP	1.25	Rs.
	IP	OP
Pathology	375	300
Blood & Componentes	2,100	
Radiology	375	300
Sonography	1,125	900
C.T. Scan	3,000	2,400
MRI	7,500	6,000
BMD	1,125	900
Mammography	2,625	2,100
ECG	375	300
Echocardiography	2,625	2,100
Physiotherapy	375	300
OPD		600
In-patient Consultation	1,200	
Endoscopy		
UGI	5,500	
LGI	6,500	
PET Scan	15,000.00	
Radiotherapy (Linec)	7,000	
Brachytherapy	3,000	
Minor OT / Biopsy per Day		
Dr's Charges	3,000	
OT Charges	1,200	
Consumables	1,000	

Major Surgeries		
Dr's Charges	30,000	
Anesthetists	10,500	
OT Charges	12,000	
Consumables	10,000	
Supramajor Surgeries		
OT Charge	36,000	
Surgeons Charge	60,000	
Anesthetist Charge	18,000	
Consumable Charges	15,000	
BMT		
Pretansplant Work up & Test	50,000	
Transplant Procedure plus Stay	12,00,000	
Post Transplant Medication	60,000	
Chemotherapy	50,000	
Admissions	100	
Pharmacy	2,400	
EHS		3,000

➤ **BED REVENUE**

Phae 1 & Phase 2 are considered together starting operations at low occupancy levels .			
ROOM REVENUE			INR
Category	Beds	Rate	Total
Pay Bed	100		
Economy Ward	20	1,200	8,760,000
Twin Sharing Room	20	2,000	14,600,000
Single Room	16	4,000	23,360,000
Deluxe Room	2	6,000	4,380,000
Suite Room	2	9,000	6,570,000
Chemotherapy Chair	8	2,000	5,840,000
Daycare	6	2,000	4,380,000
Isolation Bed	2	12,000	8,760,000
IMCU	10	6,000	21,900,000
SICU	10	6,000	21,900,000
BMT (cost will be part of the package)	4		0
			120,450,000
Service Bed	8		
Emergency	4	0	0
Preop / Post Op	4		
TOTAL BED	108		
Charity @	10%		12,045,000
GRAND TOTAL			108,405,000

➤ **DEPARTMENT REVENUE:**

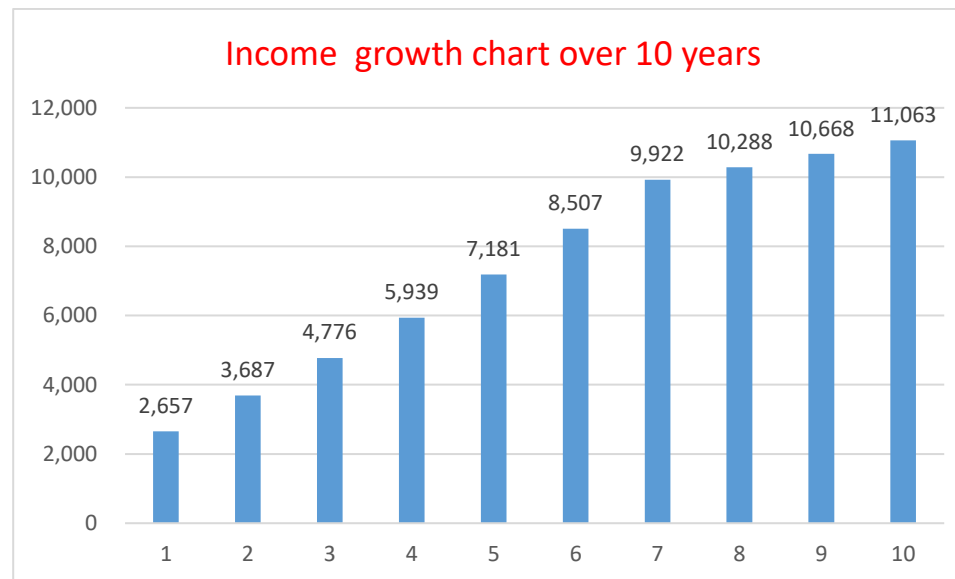
Year	1	2	3	4	5	6	7	8	9	10
Pathology	187.99	250.65	313.31	375.98	418.95	501.30	563.96	563.96	563.96	563.96
Blood & Components	45.36	60.48	60.48	75.60	75.60	90.72	90.72	105.84	105.84	120.96
Radiology	15.61	20.81	26.01	31.21	35.93	41.61	46.82	46.82	46.82	46.82
Sonography	8.17	10.89	13.61	16.34	18.65	21.78	24.50	24.50	24.50	24.50
C.T. Scan	60.84	81.12	101.40	121.68	149.86	162.24	182.52	182.52	182.52	182.52
MRI	108.90	145.20	181.50	217.80	248.64	290.40	326.70	326.70	326.70	326.70
BMD	1.97	2.63	3.29	3.94	5.75	5.26	5.91	5.91	5.91	5.91
Mammography	10.35	13.80	17.25	20.70	24.14	27.59	31.04	31.04	31.04	31.04
ECG	15.61	20.81	26.01	31.21	35.93	41.61	46.82	46.82	46.82	46.82
Echocardiography	9.53	12.71	15.88	19.06	21.76	25.41	28.59	28.59	28.59	28.59
Physiotherapy	6.94	9.25	11.56	13.87	16.35	18.50	20.81	20.81	20.81	20.81
OPD	129.60	172.80	216.00	259.20	0.00	345.60	388.80	388.80	388.80	388.80
In-patient Consultation	131.40	175.20	219.00	262.80	0.00	350.40	394.20	394.20	394.20	394.20
Endoscopy	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
UGI	9.90	13.20	16.50	19.80	0.00	26.40	29.70	29.70	29.70	29.70
LGI	11.70	15.60	19.50	23.40	0.00	31.20	35.10	35.10	35.10	35.10
PET Scan	81.00	216.00	135.00	162.00	189.00	216.00	243.00	243.00	243.00	243.00
Radiotherapy (Linec)	504.00	672.00	840.00	1,008.00	1,176.00	1,344.00	1,512.00	1,512.00	1,512.00	1,512.00
Brachytherapy	21.60	28.80	36.00	43.20	50.40	57.60	64.80	64.80	64.80	64.80
Minor OT / Biopsy	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Dr's Charges	16.20	21.60	27.00	32.40	37.80	43.20	48.60	48.60	48.60	48.60
OT Charges	6.48	8.64	10.80	12.96	15.12	17.28	19.44	19.44	19.44	19.44
Consumables	5.40	7.20	9.00	10.80	12.60	14.40	16.20	16.20	16.20	16.20
Major Surgeries	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Dr's Charges	54.00	72.00	90.00	108.00	126.00	144.00	162.00	162.00	162.00	162.00
Anesthetists	18.90	25.20	31.50	37.80	44.10	50.40	56.70	56.70	56.70	56.70
OT Charges	21.60	28.80	36.00	43.20	50.40	57.60	64.80	64.80	64.80	64.80
Consumables	18.00	24.00	30.00	36.00	42.00	48.00	54.00	54.00	54.00	54.00
Supramajor Surgeries	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

OT Charge	194.40	259.20	324.00	388.80	453.60	518.40	583.20	583.20	583.20	583.20
Surgeons Charge	324.00	432.00	540.00	648.00	756.00	864.00	972.00	972.00	972.00	972.00
Anesthetist Charge	97.20	129.60	162.00	194.40	226.80	259.20	291.60	291.60	291.60	291.60
Consumable Charges	81.00	108.00	135.00	162.00	189.00	216.00	243.00	243.00	243.00	243.00
BMT	0.00	0.00	135.00	162.00	189.00	216.00	243.00	243.00	243.00	243.00
Pretansplant Work up & Test	0.00	2.00	2.50	3.00	3.50	4.00	4.50	5.00	5.50	6.00
Transplant Procedure plus Stay	0.00	48.00	60.00	72.00	84.00	96.00	108.00	120.00	132.00	144.00
Post Transplant Medication	0.00	2.40	3.00	3.60	4.20	4.80	5.40	6.00	6.60	7.20
Chemotherapy	360.00	480.00	600.00	720.00	840.00	960.00	1,080.00	1,080.00	1,080.00	1,080.00
Pharmacy	262.80	350.40	438.00	525.60	613.20	700.80	788.40	788.40	788.40	788.40
TOTAL	2,295	3,220	4,010	4,815	4,928	6,410	7,200	7,228	7,241	7,270
Less Charity @	229	322	401	482	493	641	720	723	724	727
10%	2,065	2,898	3,609	4,334	4,435	5,769	6,480	6,505	6,517	6,543
Capacity	30%	40%	50%	60%	70%	80%	90%	90%	90%	90%
Revenue	2,065	2,898	3,768	4,702	5,705	6,781	7,934	8,251	8,581	8,924
(Plus inflation @ 4%)										

- **10% of the total income** is considered for **charity** as per the charity commissioner's norms for trust or NGO. It is deducted from the gross revenue and then the total revenue for the hospital is considered.

➤ **TOTAL INCOME OF THE HOSPITAL:**

Total income consists of bed revenue, department revenue, pharmacy with consumable revenue, apartment block rental revenue and other income. It increases every year as per the occupancy growth considered.

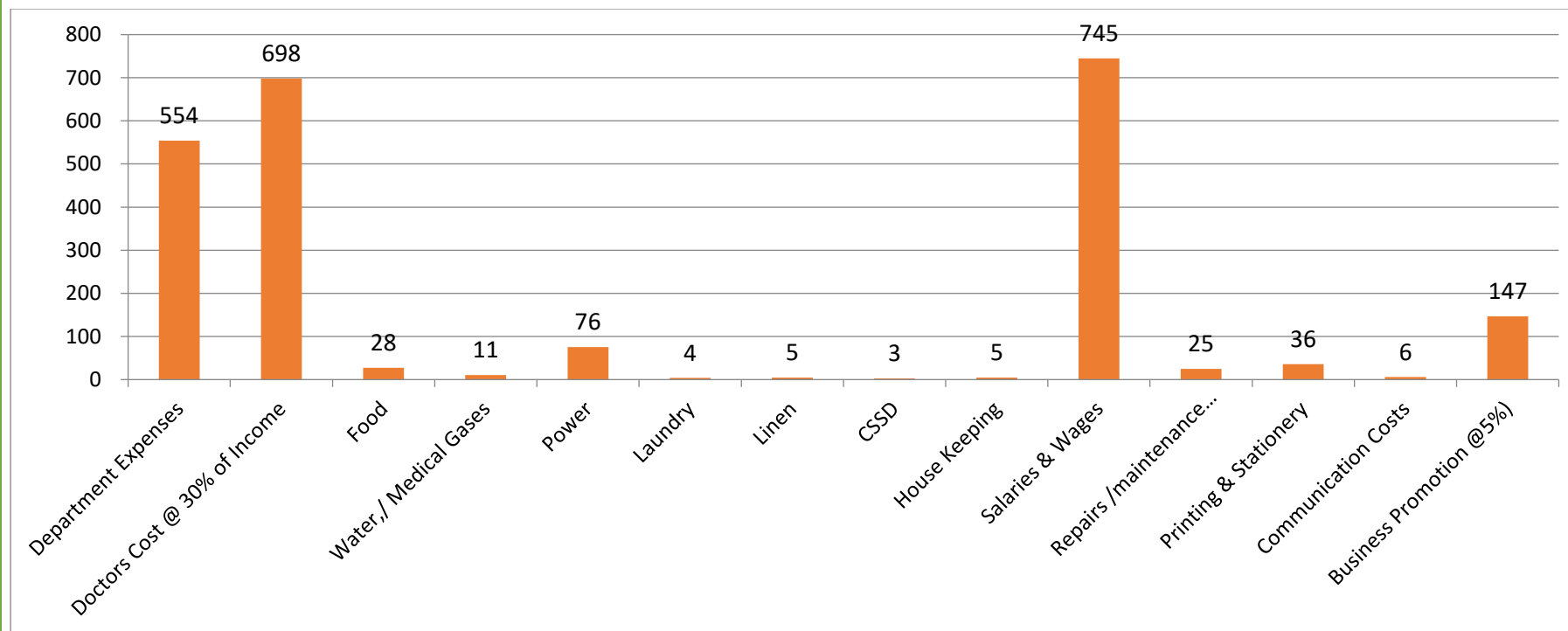


<u>INCOME</u>		Rs Lakhs >>>>>>>>>								
Year	2,026	2,027	2,028	2,029	2,030	2,031	2,032	2,033	2,034	2,035
Department Revenue	2,065	2,898	3,768	4,702	5,705	6,781	7,934	8,251	8,581	8,924
Pharmacy	263	350	438	526	613	701	788	788	788	788
Bed Revenue	325	434	564	704	854	1,015	1,187	1,235	1,284	1,335
		0	0	0	0	0	0	0	0	0
Other Income	3	5	6	8	9	11	13	14	14	15
Total a)	2,657	3,687	4,776	5,939	7,181	8,507	9,922	10,288	10,668	11,063

➤ EXPENSE ESTIMATES

Hospitals are highly technology and labor intensive with a very high operational cost. Thus it is of paramount importance to account for all possible costs from various heads so as to arrive at the right projections to understand the financial viability of the project. At the same time the correct expenditure projections and analysis helps to contain cost in future for better viability. The major expenditure heads are as follows:

EXPENSE CHART: INR LAKHS OF 1ST YEAR – EXPENSE DISTRIBUTION



➤ **Salaries & Wages**

- Cancer Hospital is a speciality hospital and speciality hospitals normally require a manpower to bed ratio of 3 : 1 to 4.5 : 1 depending on the service mix and the services required for the target clientele.
- We have kept it around 3.0 : 1. Though it seems little low but with good operational systems and computerization it is possible to manage the hospital efficiently.
- The salaries are calculated for all categories of staff.
- Calculated at 100 percent capacity and are than pegged as a percentage of the total to reach at full capacity in 7th year.
- A margin of 15% extra has been added to provide for allowances and bonus.

SALARY & WAGES		Rs. In Lakhs	
<u>Category</u>	Salary per	Number of	Total per
	month	staff	month
<i>Administration</i>			
CEO / Centre Head	450,000	1	450,000
Nursing Superintendent	120,000	1	120,000
Manager	72,000	4	288,000
Asst. Managers	42,000	4	168,000
Executives	27,000	4	108,000
Social Worker/PRO	40,000	1	40,000
Clerks /Secretary	18,000	24	432,000
Receptionists / Telephone Operator	15,000	6	90,000
Security	15,000	24	360,000
Sr. Biomedical Engineer	120,000	1	120,000
Biomedical Engineer Sr.	30,000	1	30,000
EDP Manager	60,000	1	60,000
EDP Executives	24,000	1	24,000
Sr. Engineer	60,000	1	60,000

Jr. Engineer	24,000	1	24,000
Maintenance Technicians	18,000	8	144,000
Ambulance / drivers	15,000	4	60,000
Medical			
Physician	300,000	1	300,000
Sr. Radiologist	300,000	1	300,000
Jr. Radiologist	200,000	1	200,000
Radiotherapist	300,000	1	300,000
Pathologist	150,000	1	150,000
Microbiologist	80,000	1	80,000
Intensivist	400,000	1	400,000
Anesthetist	300,000	1	300,000
Jr. Anesthetist / Intensivist	200,000	2	400,000
Dentist	50,000	1	50,000
RMOs (MBBS)	60,000	24	1,440,000
RMOs (ISM))	24,000	0	0
Nursing Supervisors	45,000	2	90,000
Incharge Nurse	36,000	4	144,000
Senior Nurse	27,000	16	432,000
Staff Nurse	18,000	84	1,512,000
Respiratory Technicians	48,000	1	48,000
Physicist	90,000	1	90,000
Senior Tech	48,000	6	288,000
Junior Tech	18,000	24	432,000
Dietician	42,000	2	84,000
Physiotherapist	24,000	2	48,000
Class Four			
Attendants	14,000	70	980,000
Total per month		334	10,646,000
Total per annum			127,752,000
Add : Margin for Bonus & Allowances excluding Doctors)----->	15.00	%	19,162,800
TOTAL			146,914,800

	Rs. Lakhs	>>>>>								
	2,026	2,027	2,028	2,029	2,030	2,031	2,032	2,033	2,034	2,035
Capacity utilization	50%	60%	70%	80%	90%	100%	100%	100%	100%	100%
Expense at stated capacity	734.57	881.49	1,028.40	1,222	1,430	1,653	1,719	1,787	1,859	1,933

(Plus inflation @ 4% from 3rd year onwards)

➤ **DEPARTMENT EXPENSES**

Based on the industry standards and volumes, the departmental expenses are estimated for salary and consumables as a direct percentage of the gross revenue generated from the departments.

- Doctor's share is in addition to the salaries of certain doctors which is part of total salary. It is considered at 30% of the gross revenue and is in line with industry standards.

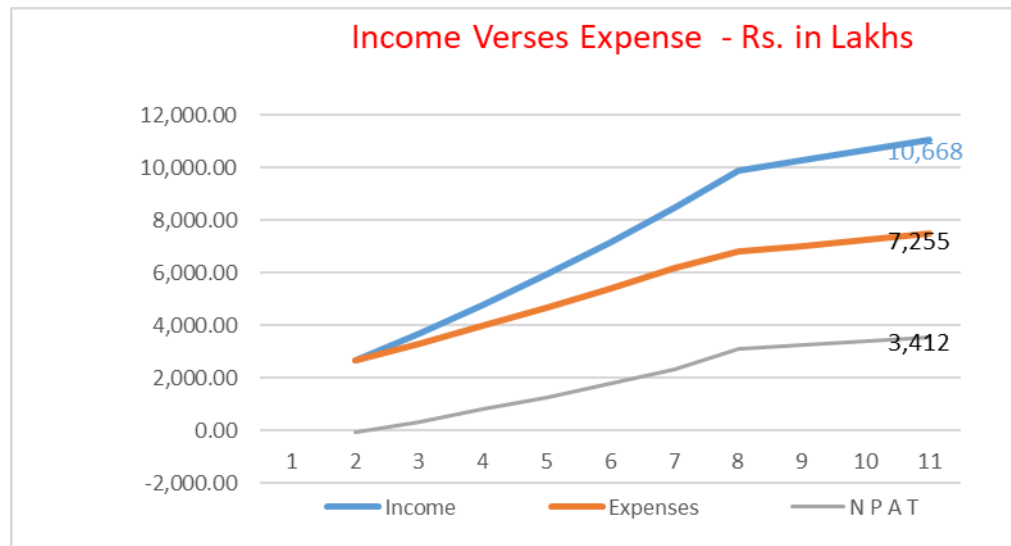
Expense			Total Expenses									
	Consumable	Dr's	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10
Pathology	20%	2%	41.36	55.14	68.93	82.71	92.17	110.29	124.07	124.07	124.07	124.07
Blood & Componentets	20%	2%	9.98	13.31	13.31	16.63	16.63	19.96	19.96	22.98	23.28	23.28
Radiology	20%	2%	3.43	4.58	5.72	6.87	7.91	9.15	10.30	10.30	10.30	10.30
Sonography	10%	2%	0.98	1.31	1.63	1.96	2.24	2.61	2.94	2.94	2.94	2.94
C.T. Scan	10%	2%	7.30	9.73	12.17	14.60	17.98	19.47	21.90	21.90	21.90	21.90
MRI	10%	2%	13.07	17.42	21.78	26.14	29.84	34.85	39.20	39.20	39.20	39.20
BMD	10%	2%	0.24	0.32	0.39	0.47	0.69	0.63	0.71	0.71	0.71	0.71
Mammography	10%	2%	1.24	1.66	2.07	2.48	2.90	3.31	3.73	3.73	3.73	3.73
ECG	5%	2%	1.09	1.46	1.82	2.18	2.52	2.91	3.28	3.28	3.28	3.28
Echocardiography	10%	2%	1.14	1.52	1.91	2.29	2.61	3.05	3.43	3.43	3.43	3.43
Physiotherapy	5%	2%	0.49	0.65	0.81	0.97	1.14	1.29	1.46	1.46	1.46	1.46
OPD			0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
In-patient Consultation	0%		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Endoscopy			0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
UGI	5%		0.50	0.66	0.83	0.99	0.00	1.32	1.49	1.49	1.49	1.49
LGI	5%		0.59	0.78	0.98	1.17	0.00	1.56	1.76	1.76	1.76	1.76
PET Scan	25%		20.25	54.00	33.75	40.50	47.25	54.00	60.75	60.75	60.75	60.75
Radiotherapy (Lineec)	5%		25.20	33.60	42.00	50.40	58.80	67.20	75.60	75.60	75.60	75.60
Brachytherapy	5%		1.08	1.44	1.80	2.16	2.52	2.88	3.24	3.24	3.24	3.24
Minor OT / Biopsy per Day			0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Dr's Charges			0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
OT Charges	5%		0.32	0.43	0.54	0.65	0.76	0.86	0.97	0.97	0.97	0.97
Consumables	5%		0.27	0.36	0.45	0.54	0.63	0.72	0.81	0.81	0.81	0.81
Major Surgeries			0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Dr's Charges			0.00	0.00	0.00							
Anesthetists			0.00	0.00	0.00							
OT Charges	5%		1.08	0.00	0.00							
Consumables	60%		10.80	14.40	18.00	21.60	25.20	28.80	32.40	32.40	32.40	32.40
Supramajor Surgeries			0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
OT Charge	5%		9.72	12.96	16.20	19.44	22.68	25.92	29.16	29.16	29.16	29.16
Surgeons Charge			0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Anesthetist Charge		5%	4.86	6.48	8.10	9.72	11.34	12.96	14.58	14.58	14.58	14.58
Consumable Charges	60%	5%	52.65	70.20	87.75	105.30	122.85	140.40	157.95	157.95	157.95	157.95
BMT			0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Pretansplant Work up &	10%		0.00	0.20	0.25	0.30	0.35	0.40	0.45	0.55	0.55	0.55
Transplant Procedure &	10%		0.00	4.80	6.00	7.20	8.40	9.60	10.80	13.20	13.20	13.20
Post Transplant Medical	60%		0.00	1.44	1.80	2.16	2.52	2.88	3.24	3.96	3.96	3.96
Chemotherapy	50%		180.00	240.00	300.00	360.00	420.00	480.00	540.00	540.00	540.00	540.00
Pharmacy	50%		448.30									

➤ Other Expenses

The other expenses have been calculated taking into account all possible contingencies and have been suitably adjusted for inflation each year. Expenses that have a direct relationship with Fixed Assets, such as depreciation and repairs & maintenance have been calculated keeping this in mind.

➤ PROFIT & LOSS

The project seems to be financially very sound. It is making book profit from year two. Cash profit starts from year one only. But this has to be understood well in terms of its dependency on the other sources of income in initial years.



PROFIT & LOSS										
INCOME		Rs Lakhs >>>>>>>								
Year	2,026	2,027	2,028	2,029	2,030	2,031	2,032	2,033	2,034	2,035
Department Revenue	2,065	2,898	3,768	4,702	5,705	6,781	7,934	8,251	8,581	8,924
Pharmacy	263	350	438	526	613	701	788	788	788	788
Bed Revenue	325	434	564	704	854	1,015	1,187	1,235	1,284	1,335
		0	0	0	0	0	0	0	0	0
Other Income	3	5	6	8	9	11	13	14	14	15
Total a)	2,657	3,687	4,776	5,939	7,181	8,507	9,922	10,288	10,668	11,063
EXPENSES										
Department Expenses	388	549	649	810	983	1,168	1,367	1,421	1,478	1,537
Doctors Cost @ 30% of Income	717	1,000	1,299	1,622	1,968	2,339	2,736	2,846	2,959	3,078
Food	22	29	37	44	51	59	68	78	89	103
Water,/ Medical Gases	11	14	19	23	28	34	39	41	43	44
Power	43	52	63	75	87	101	105	109	114	118
Laundry	3	4	5	7	8	9	10	10	10	10
Linen	5	7	8	9	11	13	13	14	14	15
CSSD	2	3	4	5	6	7	8	8	9	9
House Keeping	4	4	5	6	7	9	9	9	10	10
Communication Costs	6	6	7	7	7	8	8	8	9	9
Business Promotion @4%	106	112	117	123	129	136	142	150	157	165
Salaries & Wages	735	881	1,028	1,222	1,430	1,653	1,719	1,787	1,859	1,933
Repairs /maintenance /upg	0	5	52	104	114	126	138	152	168	184
Printing & Stationery	18	19	20	21	22	23	24	25	27	28
Miscellaneous	41	54	66	82	97	114	128	133	139	145
Insurance	38	40	43	46	49	53	57	61	65	69
Legal	6	6	7	7	7	8	8	8	9	9
Audit	6	6	7	7	7	8	8	8	9	9
Sub - Total b)	2,151	2,792	3,436	4,219	5,013	5,864	6,587	6,870	7,166	7,477
PBDIT a-b	506	895	1,340	1,719	2,168	2,644	3,335	3,418	3,501	3,586
Depreciation	783	673	657	581	506	442	388	342	302	269
Preliminary Exp written Off	8	8	8	8	8	8	8	8	8	8
PBIT	-285	214	675	1,131	1,654	2,194	2,939	3,068	3,192	3,310
Interest	539	539	537	463	388	313	239	164	89	15
Net Surplus Before Tax	-824	-325	138	668	1,266	1,880	2,701	2,904	3,102	3,295
Less: Tax	0	0	6	132	29	0	0	0	0	0
N P A T	-824	-325	132	638	1,237	1,880	2,701	2,904	3,102	3,295

BALANCE SHEET											
Rs lakhs >>>>>											
Year	Constn	2,026	2,027	2,028	2,029	2,030	2,031	2,032	2,033	2,034	2,035
<u>SOURCES OF FUNDS</u>											
Promoters Contribution	3,783.50	3,783.50	3,783.50	3,783.50	3,783.50	3,783.50	3,783.50	3,783.50	3,783.50	3,783.50	3,783.50
Credit party Deposits		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Reserves & surplus		-824.28	-1,149.22	-1,017.51	-379.65	857.60	2,737.97	5,438.75	8,343.12	11,445.36	14,740.53
Secured loans	5,675.25	5,675.25	5,675.25	5,657.24	4,871.01	4,084.79	3,298.56	2,512.33	1,726.10	939.87	153.64
T O T A L	9,459	8,634	8,310	8,423	8,275	8,726	9,820	11,735	13,853	16,169	18,678
<u>APPLICATIONS OF FUNDS</u>											
Land	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00
Fixed Assets	7,817.53	7,817.53	7,034.35	7,641.81	6,984.90	6,403.64	5,897.31	5,454.82	5,066.70	4,724.91	4,422.62
less : Depreciation		783.18	672.90	656.91	581.27	506.32	442.49	388.12	341.79	302.29	268.60
a >	7,817.53	7,034.35	6,361.45	6,984.90	6,403.64	5,897.31	5,454.82	5,066.70	4,724.91	4,422.62	4,154.02
Preliminary Expenses	75.43	75.43	67.89	60.35	52.80	45.26	37.72	30.17	22.63	15.09	7.54
less : Written off		7.54	7.54	7.54	7.54	7.54	7.54	7.54	7.54	7.54	7.54
b >	75.43	67.89	60.35	52.80	45.26	37.72	30.17	22.63	15.09	7.54	-0.00
Current Assets											
Deposits			0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Short term invest	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Cash & Bank	1,065.79	1,207.34	1,616.01	1,167.14	1,672.58	2,703.28	4,317.99	6,688.12	9,178.76	11,828.87	14,639.39
i >	1,065.79	1,207.34	1,616.01	1,167.14	1,672.58	2,703.28	4,317.99	6,688.12	9,178.76	11,828.87	14,639.39
Less :											
Current Liabilities											
Bank Overdraft		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Provisions & Creditors		175.11	228.27	281.61	346.61	412.42	482.96	542.87	566.04	590.30	615.74
ii >	0.00	175.11	228.27	281.61	346.61	412.42	482.96	542.87	566.04	590.30	615.74
53.16											
Net Current Assets (i-ii)											
c >	1,065.79	1,032.23	1,387.74	885.52	1,325.97	2,290.86	3,835.04	6,145.25	8,612.72	11,238.57	14,023.65
T O T A L (a1+a + b + c)	9,459	8,634	8,310	8,423	8,275	8,726	9,820	11,735	13,853	16,169	18,678

CASH FLOW STATEMENT

<u>SOURCES OF FUNDS</u>	Const	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10
OPENING CASH/BK BALANCE	0	1,066	1,207	1,616	1,167	1,673	2,703	4,318	6,688	9,179	11,829
NP		-824	-325	132	638	1,237	1,880	2,701	2,904	3,102	3,295
Depreciation	0	783	673	657	581	506	442	388	342	302	269
Prel Exp written off	0	8	8	8	8	8	8	8	8	8	8
Promoters Contribution	3,783		0	0	0	0	0	0	0	0	0
Deposits	0	0	0	0	0	0	0	0	0	0	0
Term Loans - taken/(paid)	5,675		0			0	0		0	0	0
Net Current Assets - (Inc)/Dec	0	175.11	53	53	65	66	71	60	23	24	25

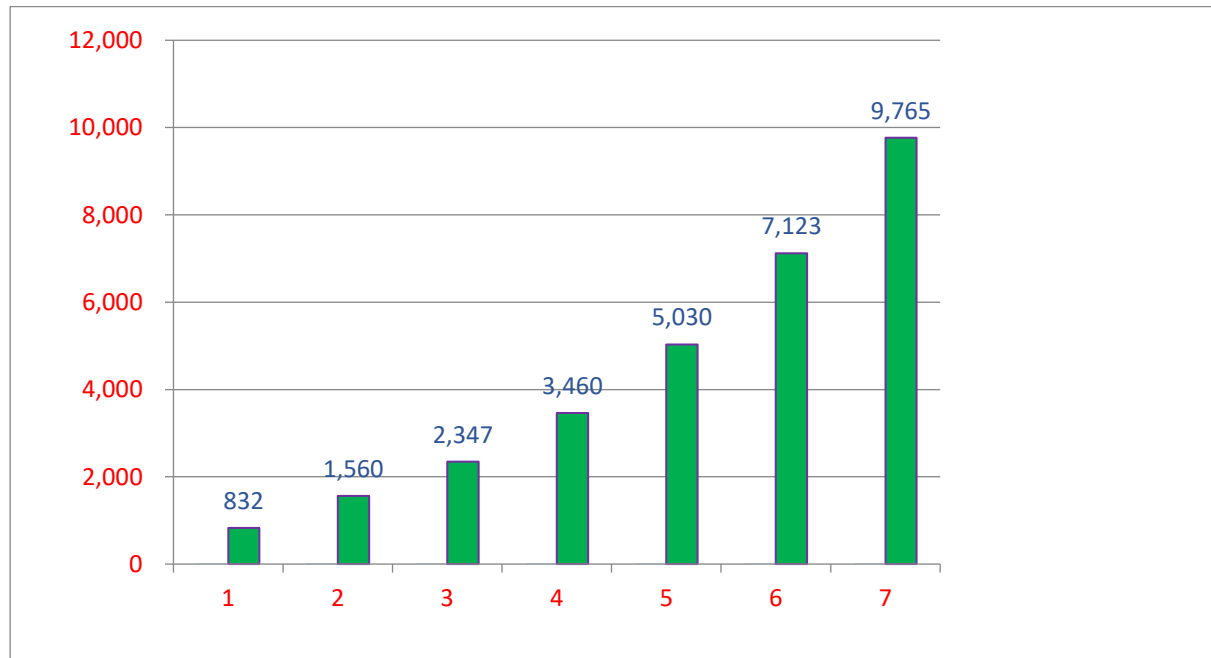
(not including Cash/Bk)

T O T A L	9,459	1,207	1,616	2,466	2,459	3,490	5,104	7,474	9,965	12,615	15,426
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APPLICATIONS OF FUNDS

Prel Exp Paid	75										
Fixed Assets bought	8,318			512							
Repayment of Term Loan		0	0	786	786	786	786	786	786	786	786
Dividends paid		0	0	0	0	0	0	0	0	0	0
	8,393	0	0	1,298	786	786	786	786	786	786	786
Add: Closing CASH/BK BALANCE	1,066	1,207	1,616	1,167	1,673	2,703	4,318	6,688	9,179	11,829	14,639
T O T A L	9,459	1,207	1,616	2,466	2,459	3,490	5,104	7,474	9,965	12,615	15,426

CASH FLOW GROWTH – 1ST TO 7TH YEAR – INR LAKHS



DSCR CALCULATIONS

	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7
NP	(824.3)	(324.9)	131.7	637.9	1,237.3	1,880.4	2,700.8
Depreciation	783.18	672.90	656.91	581.27	506.32	442.49	388.12
Interest	539.15	539.15	537.44	462.75	388.05	313.36	238.67
	498.04	887.11	1,326.06	1,681.87	2,131.63	2,636.23	3,327.57
Loan Repayment	0.00	0.00	786.23	786.23	786.23	786.23	786.23
Interest	539.15	539.15	537.44	462.75	388.05	313.36	238.67
	539.15	539.15	1,323.67	1,248.97	1,174.28	1,099.59	1,024.90
DSCR	0.92	1.65	1.00	1.35	1.82	2.40	3.25
Average DSCR	1.77						

DEBT SERVICE COVERAGE RATIO:

This is a calculation of the total cash flow in the project with relation to its capacity to service the loan. Normally a ratio of over 1:1.50 is considered good. In the proposed project it is working out to be about 2.39 which is very good, the reason being conceptualization of highly customized low cost high quality project

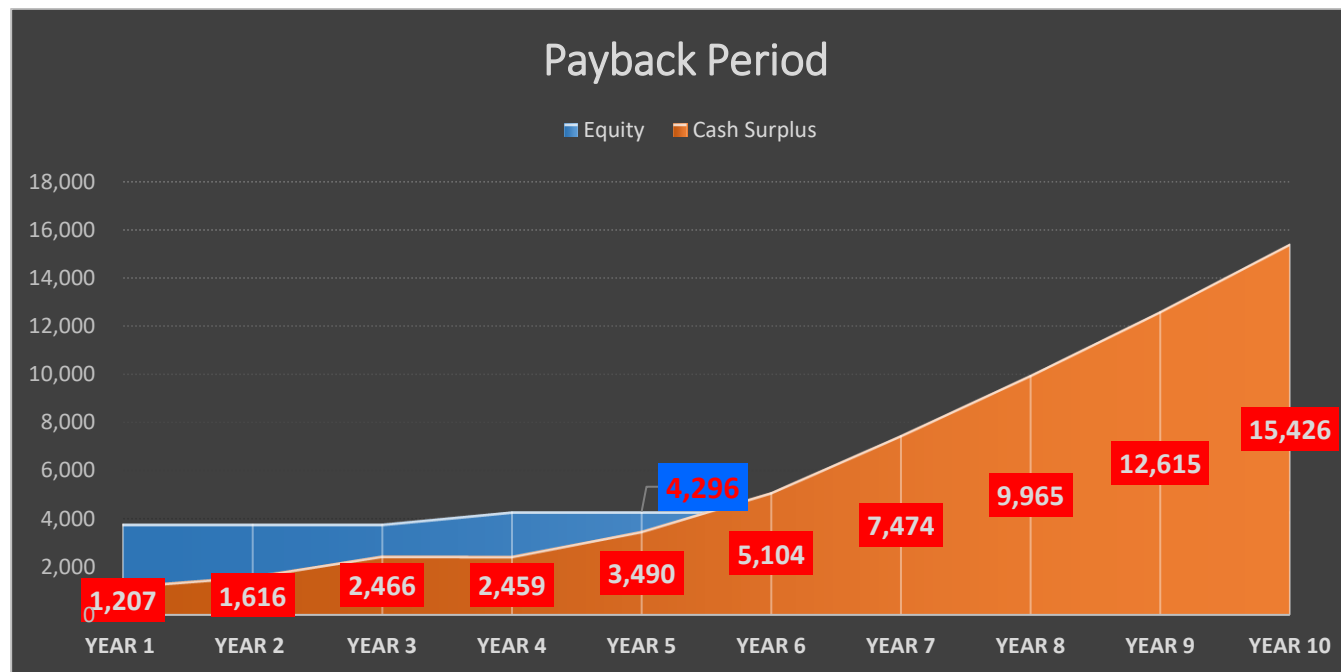
INTERNAL RATE OF RETURN:

This gives an idea in terms of the kind of return expected from a project, which helps in making investment decision. Based on the profit generated the project is showing return of over 9% which is good for healthcare charitable organizations.

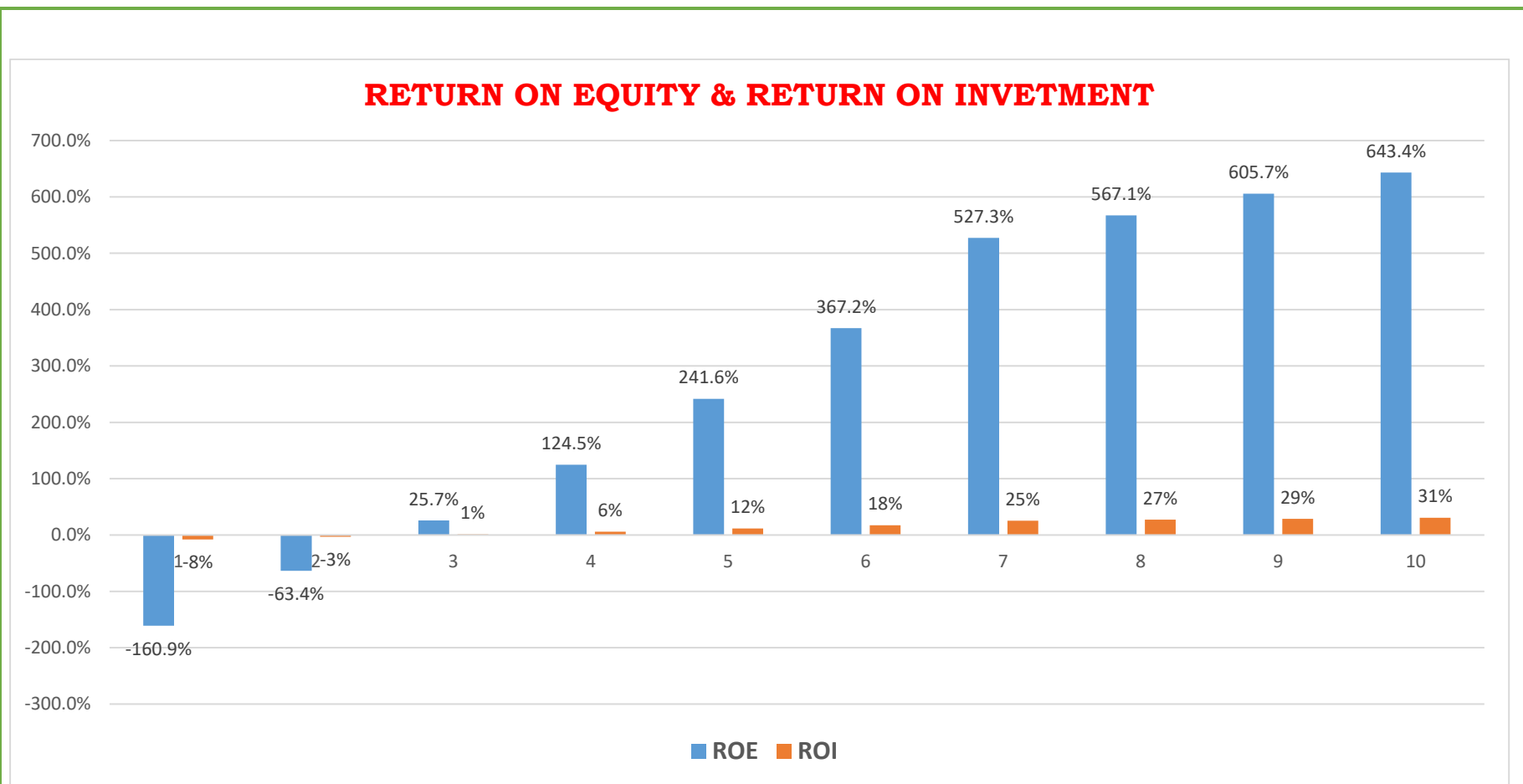
IRR

		Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7
Outflow								
Fixed Assets bought	8,318	0	0	512	0	0	0	0
Prel Exp Paid	75.43							
	8,393	0	0	512	0	0	0	0
Inflow								
NP before dep	0.00	(33.56)	355.51	796.16	1,226.67	1,751.12	2,330.41	3,096.44
Salvage Value								5,190.52
	0	(34)	356	796	1,227	1,751	2,330	8,286.96
Net flow	(8,393)	(34)	356	284	1,227	1,751	2,330	8,287

IRR	9.05%
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CAPITAL INVESTMENT					Rs. 5492.39 Lakh					
	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10
Equity	3,783	3,783	3,783	4,296	4,296	4,296	4,296	4,296	4,296	4,296
Cash Surplus	1,207	1,616	2,466	2,459	3,490	5,104	7,474	9,965	12,615	15,426
PAYBACK PERIOD					Six Years					



All the projections are done based on a good market research with detailed technical and financial analysis. Looking at the total financial calculations, it can be assumed that project is definitely financially viable.